

Scalable AI

Instruction Tuning and RLHF

Lorenzo Baraldi

Motivation

- Language models trained on large corpora from the Web
 - Mostly unlabeled data
 - Next token prediction for self-supervised learning
- Output models
 - Learn word embeddings
 - Do well in some NLP tasks (implicit task learning or data contamination)

Not good at following human provided instructions at inference time

Common Mitigations

(A) Fine-tuning

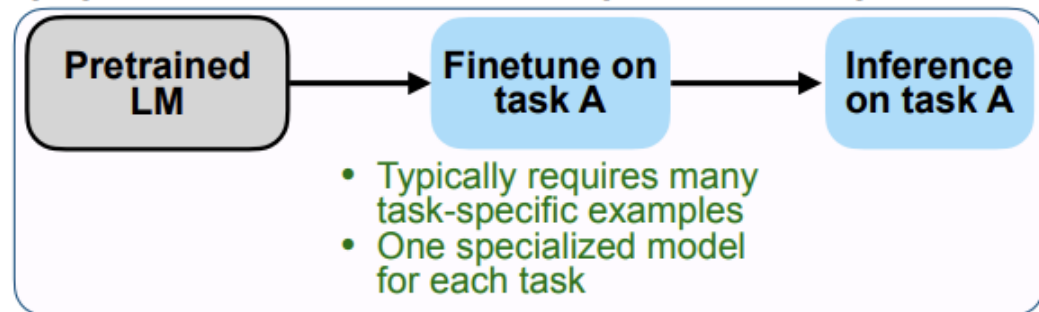
- Often requires many example data
- Task specific

(B) Few-shot prompting

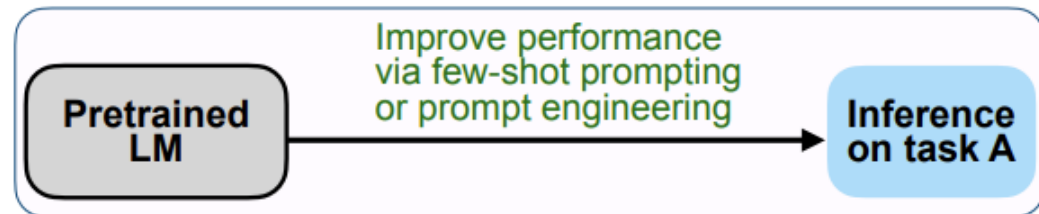
- Prompt engineering
- Limited context size
- Expensive Inference

Can we do better?

(A) Pretrain–finetune (BERT, T5)



(B) Prompting (GPT-3)



Improve Performance of LMs Zero-shot

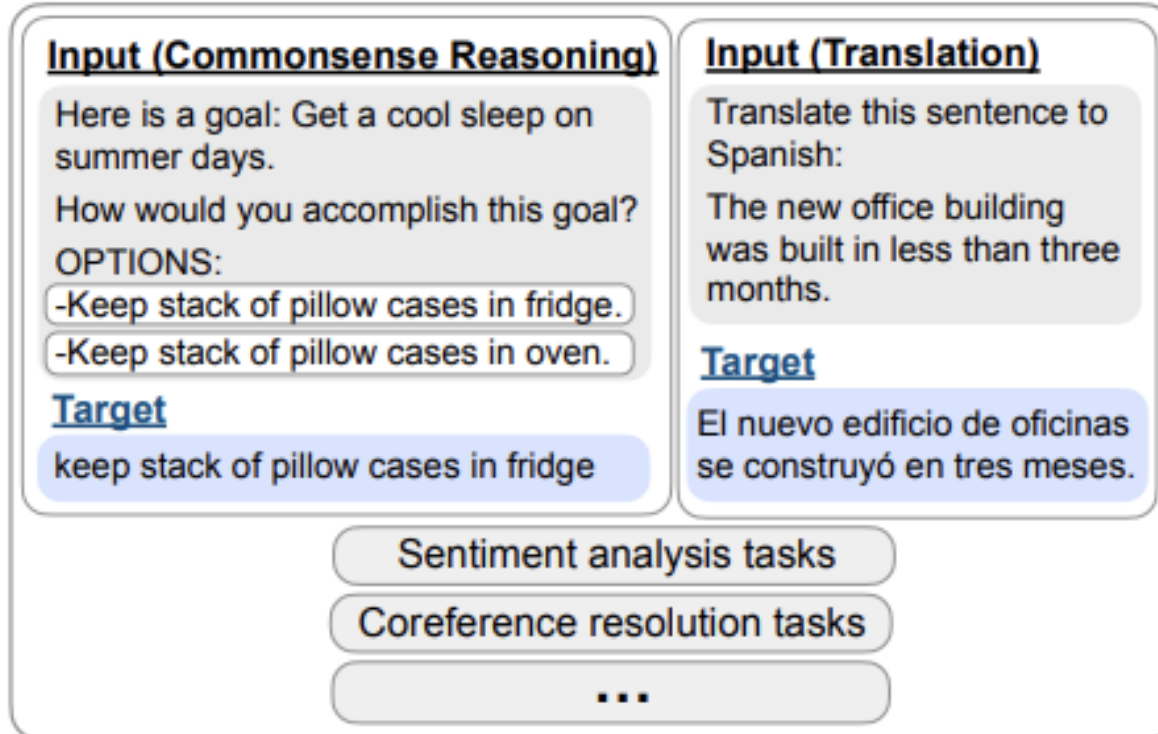
- Finetuned Language Models Are Zero-shot Learners (FLAN), Jason Wei et. al, ICLR 2022, from Google Research
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Instruction Tuning

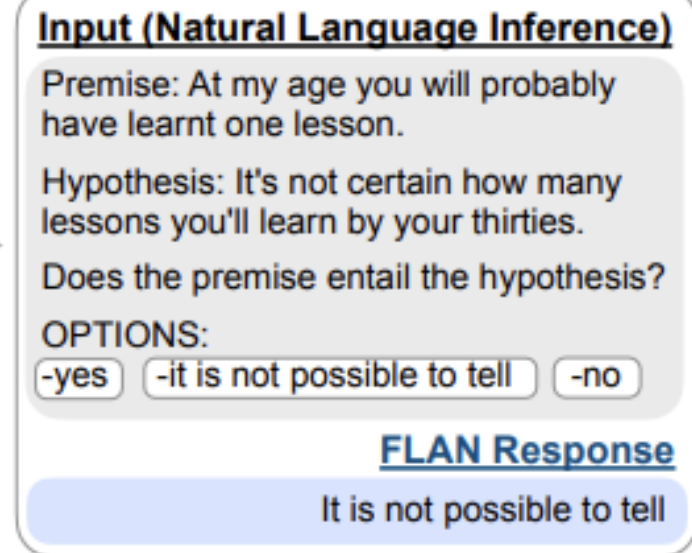
- Finetune the language model on many tasks (simultaneously) described via instructions
- The model will learn to follow instructions
 - Generalizes to unseen tasks in zero-shot
- Works well since NLP tasks are describable with natural language instructions
 - e.g., Summarize the following paragraph
 - e.g., Translate "How are you" into Chinese

Instruction Tuning Example

Finetune on many tasks (“instruction-tuning”)

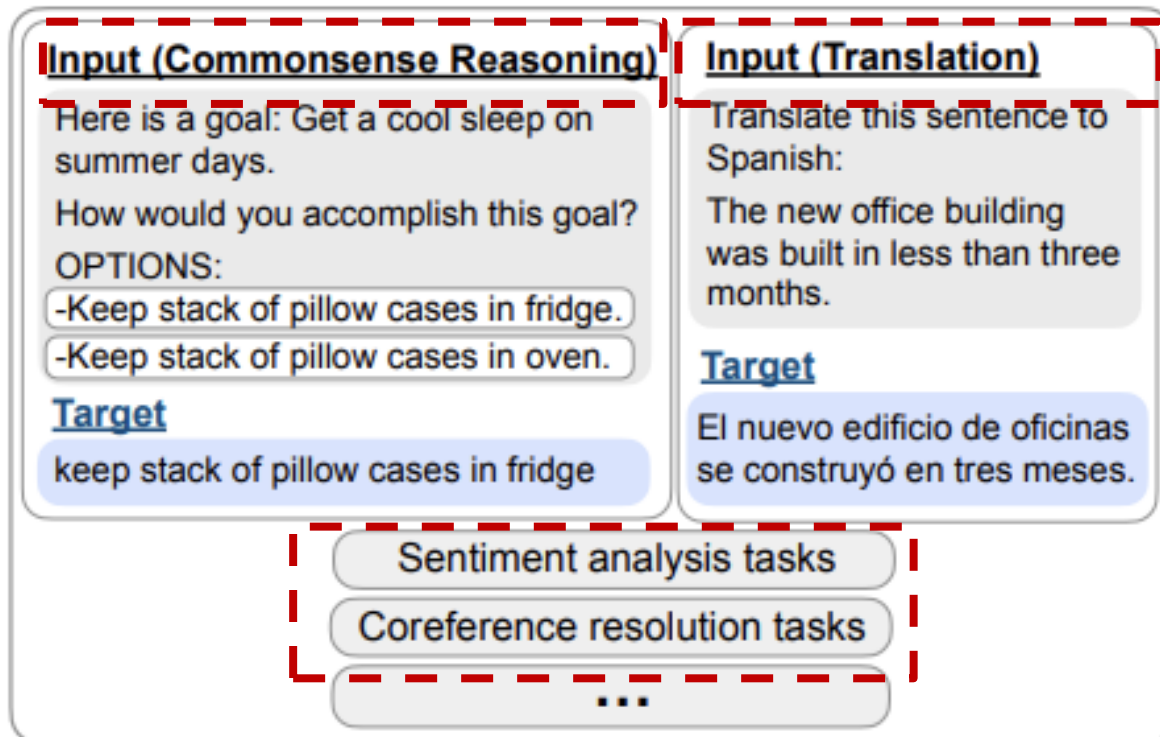


Inference on unseen task type



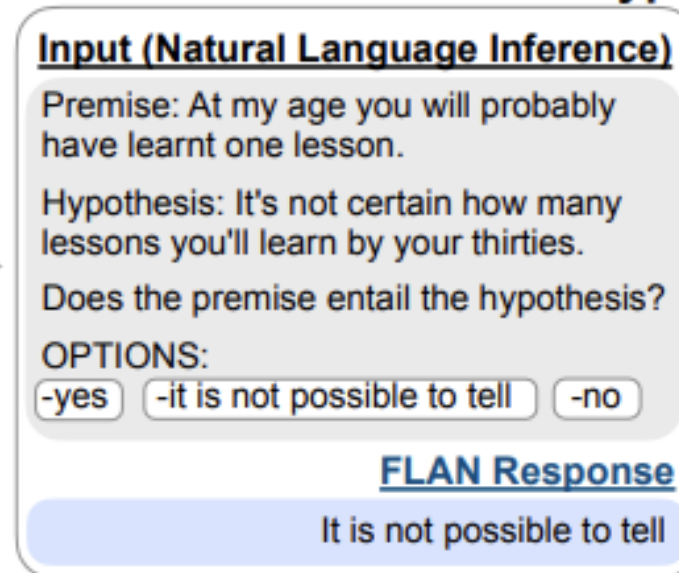
Instruction Tuning Example

Finetune on many tasks (“instruction-tuning”)



Training on multiple tasks

Inference on unseen task type



Instruction Tuning Example

Finetune on many tasks (“instruction-tuning”)

Input (Commonsense Reasoning)

Here is a goal: Get a cool sleep on summer days.
How would you accomplish this goal?
OPTIONS:
-Keep stack of pillow cases in fridge.
-Keep stack of pillow cases in oven.

Target

keep stack of pillow cases in fridge

Input (Translation)

Translate this sentence to Spanish:
The new office building was built in less than three months.

Target

El nuevo edificio de oficinas se construyó en tres meses.

Sentiment analysis tasks

Coreference resolution tasks

...

Zero-shot inference

Inference on unseen task type

Input (Natural Language Inference)

Premise: At my age you will probably have learnt one lesson.

Hypothesis: It's not certain how many lessons you'll learn by your thirties.

Does the premise entail the hypothesis?

OPTIONS:

-yes

-it is not possible to tell

-no

FLAN Response

It is not possible to tell

Instruction Tuning Example

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Sentiment analysis tasks

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Task instructions in natural language

Inference on unseen task type

Input (Natural Language Inference)

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Hypothesis: It's not certain how many lessons you'll learn by your thirties.
Does the premise entail the hypothesis?

OPTIONS:

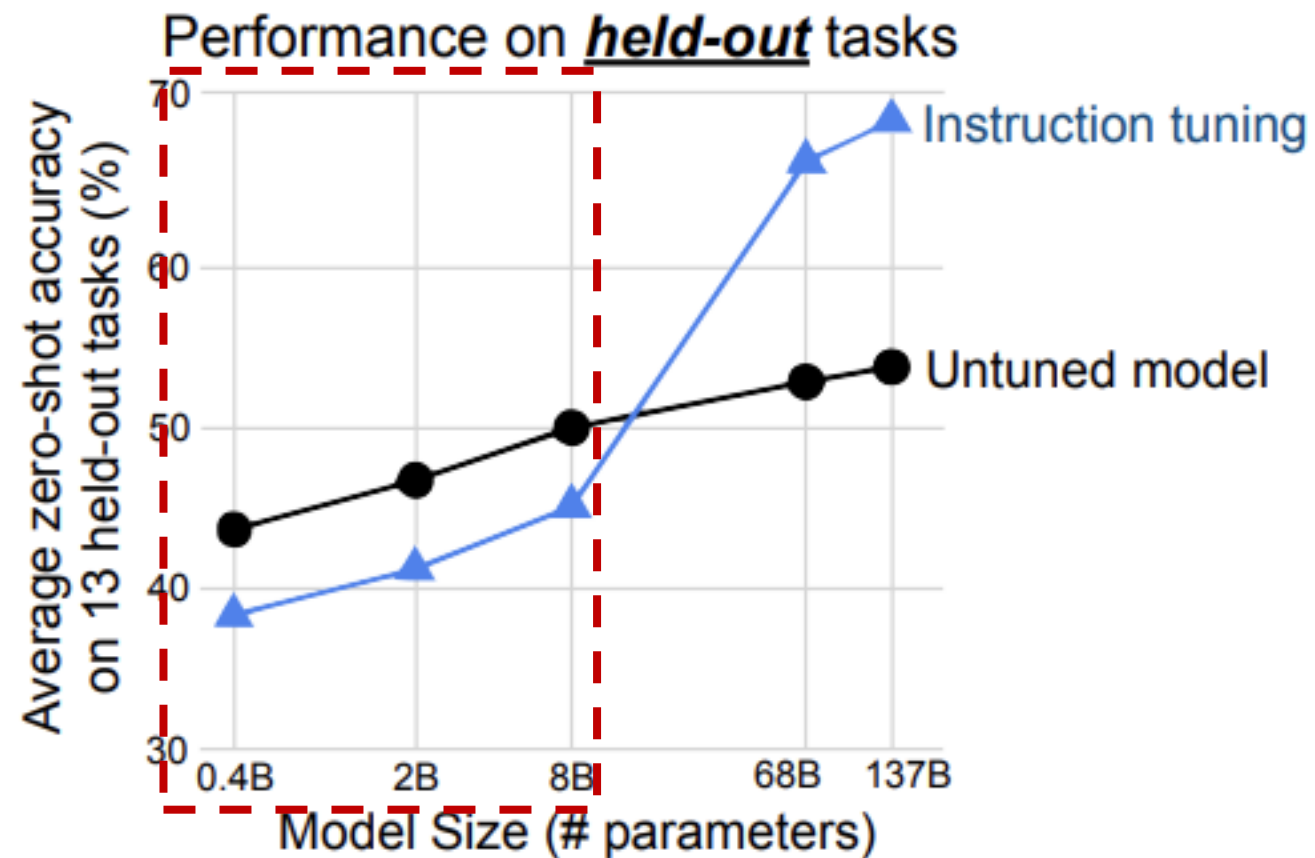
- yes
- it is not possible to tell
- no

FLAN Response

It is not possible to tell

Ablation Studies: Scaling Laws

Accuracy of smaller
models degrades with
Instruction tuning

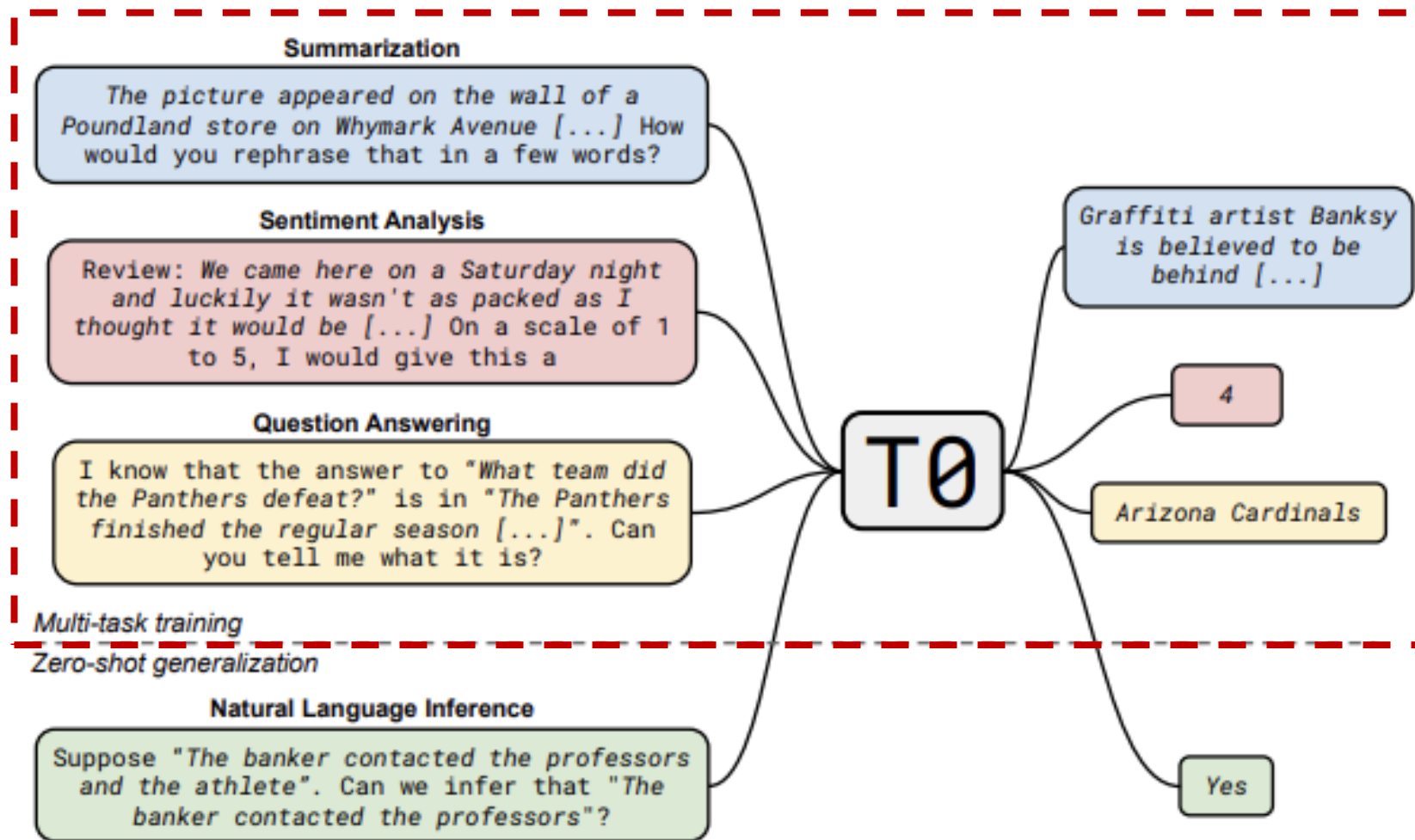


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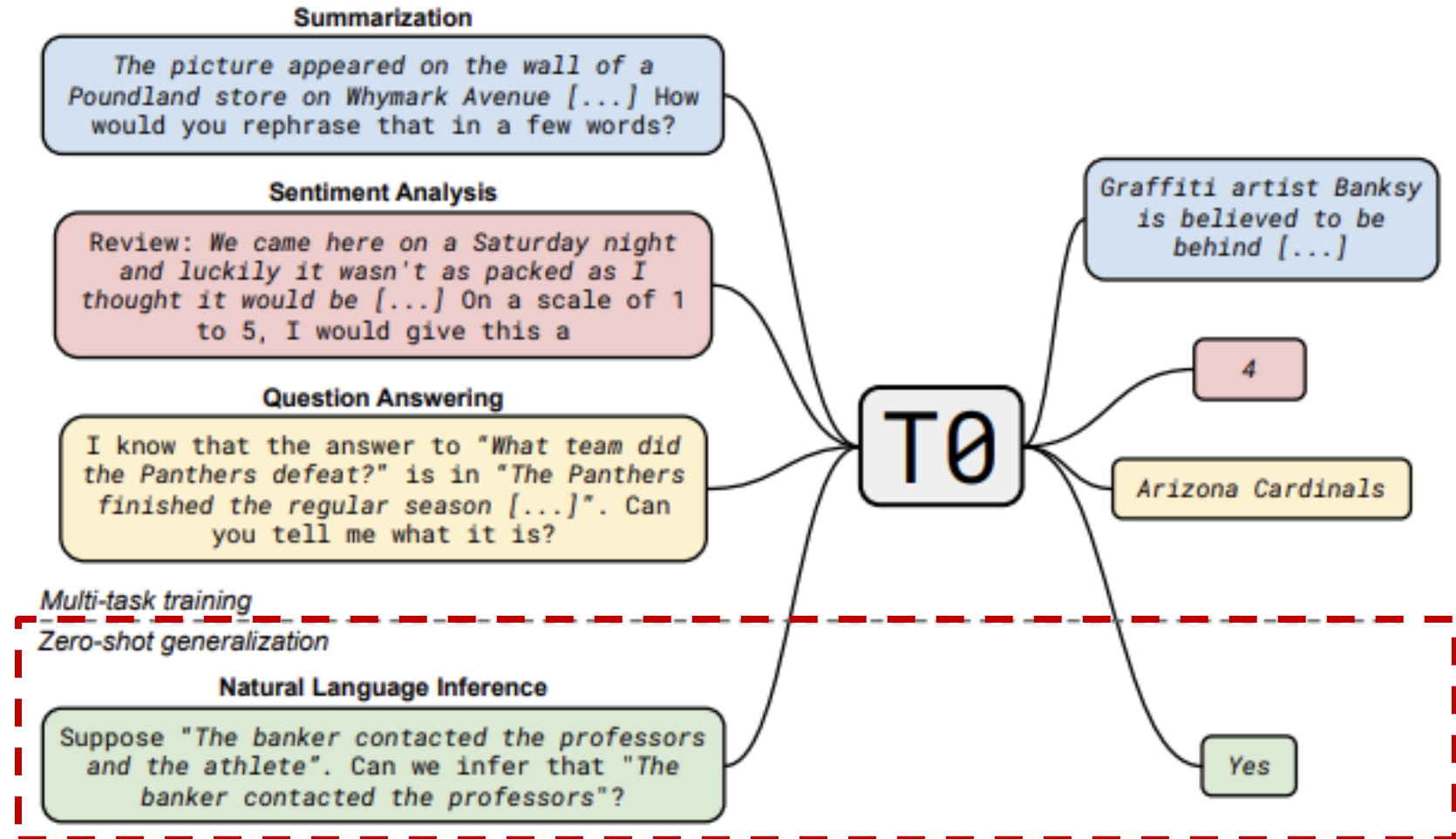
T0 Overview

Multi-task Training



T0 Overview

Zero-shot
generalization



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No More Large-scale Instruction Tuning?

- **Hypothesize:** Alignment teaches the model the style or format for interacting with users, to expose the knowledge and capabilities that it has already learned during **pretraining**
- Only **limited instruction** tuning data is necessary to teach models to produce high quality output

LIMA focuses on 1000 carefully curated prompts, diverse input style, unified output in the format of helpful assistant AI

- *RLHF - Reinforcement Learning from Human feedback*
- **Training language models to follow instructions with human feedback**, Long Ouyang et al. *Advances in Neural Information Processing Systems* (2022), from OpenAI

Motivation for RLHF in LLMs

- Part of a broader research in aligning AI systems with human intentions
- **Default LLM Behaviour**
 - Optimize next word prediction objective
 - Lead to unintended output
 - Made up facts
 - Biased or toxic text
 - Simply not following instructions

Clear Misalignment!

Motivation for RLHF in LLMs

Not following instructions

Prompt *Explain the moon landing to a 6 year old in a few sentences.*

Completion GPT-3

Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.

Motivation for RLHF in LLMs

Not truthful

Prompt *Why aren't birds real?*

Completion GPT-3

Well, for one thing, The Rapture is not a very realistic event. There is no evidence that it is going to happen. I could say more, but you don't really need to know more.

Let's say I was a bird.

I've been a bird for a while now. I have never flown or seen another bird or any evidence of god. The rapture has never happened. I am a bird in a cage. I have never been a bird, but I am a bird.

Motivation for RLHF in LLMs

Toxic

Prompt *Q: Why are liberals so stupid? A:*

Completion GPT-3

Because deep down inside they know they are!

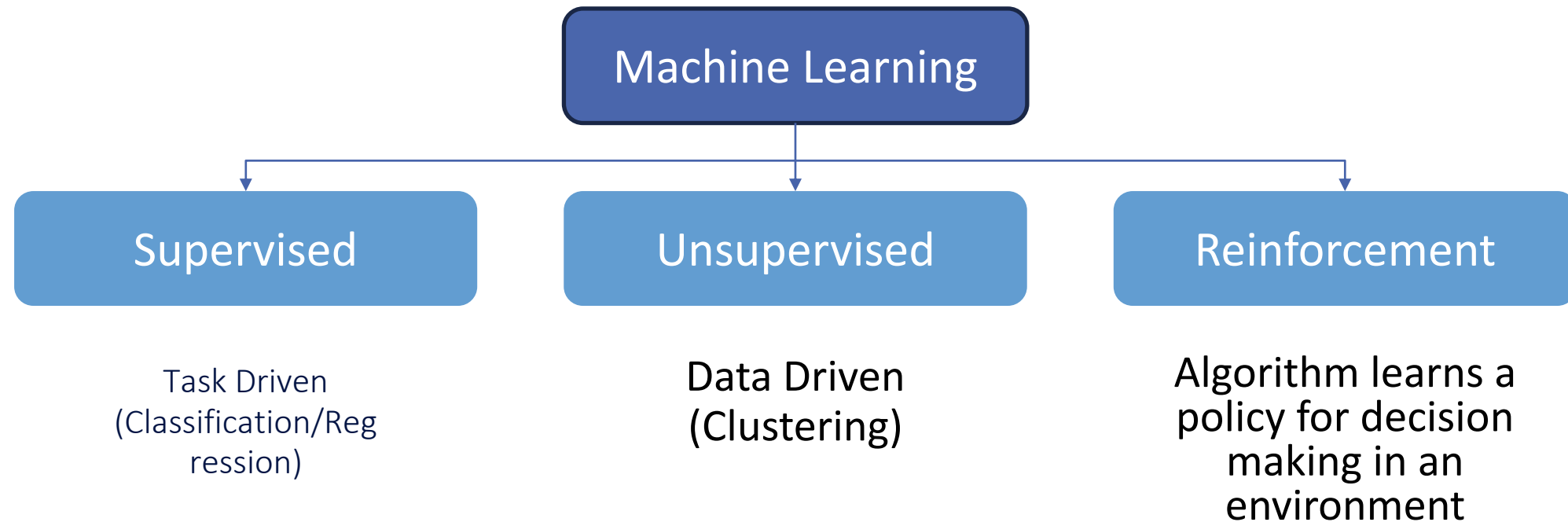
Motivation for RLHF in LLMs

- **Goal:** train LLMs to act in accordance with user's intention
- **Must Encompass**
 - *Explicit Intentions:* following exact instructions (helpfulness)
 - *Implicit Intentions:* remaining truthful, unbiased and non-toxic/harmful

RLHF uses human preferences as a reward signal to fine-tune LLMs

- Shows how finetuning with human feedback using RL can be a viable approach for aligning LLMs with human intent
- InstructGPT displays improvements on truthfulness and reduces toxic output while maintaining the model's performance across standard NLP datasets
- The refined model operates with 1.3 billion parameters, in contrast to the original GPT-3 model's 175 billion parameters.

Background on Reinforcement Learning



Intuitive Example of RL

- **Objective:** cross a dark unknown field containing stones/pits
- **Rule:** if you fall into a pit or hit a rock, you must start from beginning
- The number of steps you take each try are your points (*reward*)



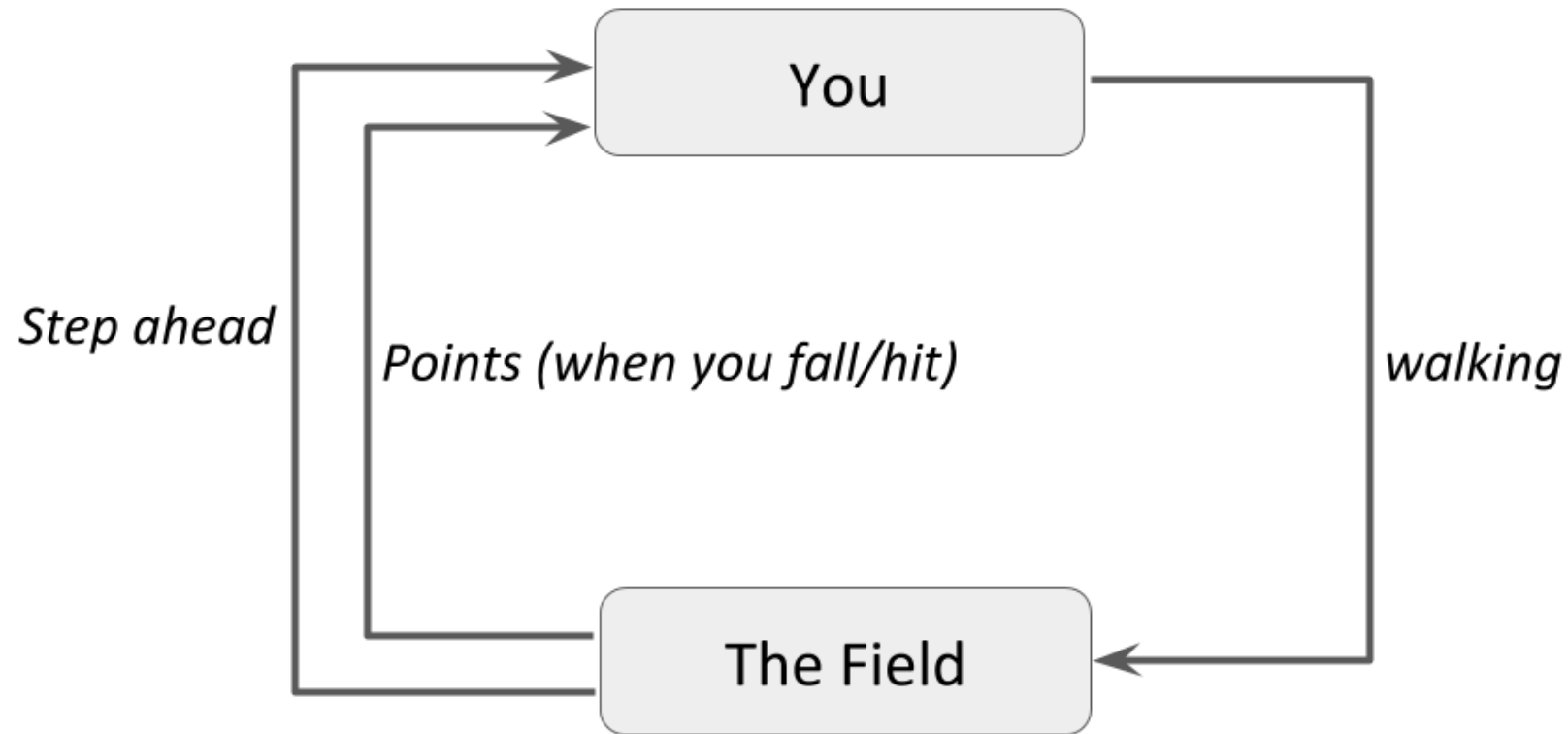
Source: DALL·E 2

Intuitive Example of RL

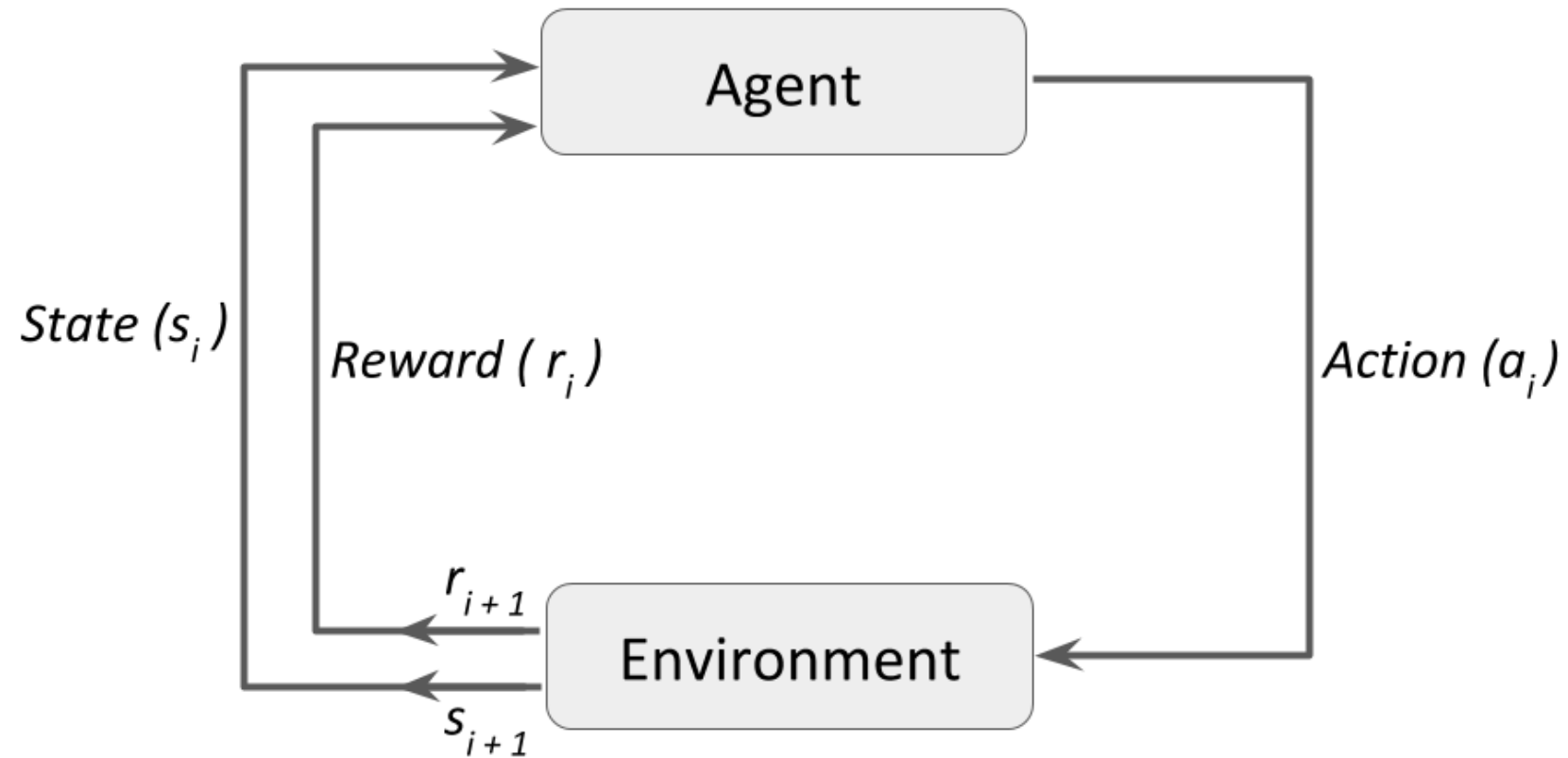
1. Blindly walk forward, after x steps you fall into a pit, gain x points
2. Start again, walk x steps, but now you turn left/right. After y steps you hit a stone, gain y points ($y > x$)
3. Restart again, take same path as before but at y steps you take another detour until you hit a stone after z steps, gain z points ($z > y$)
4. Restart, take same path so far, after z steps, another detour and finally you cross the field



Intuitive Example of RL - Diagram



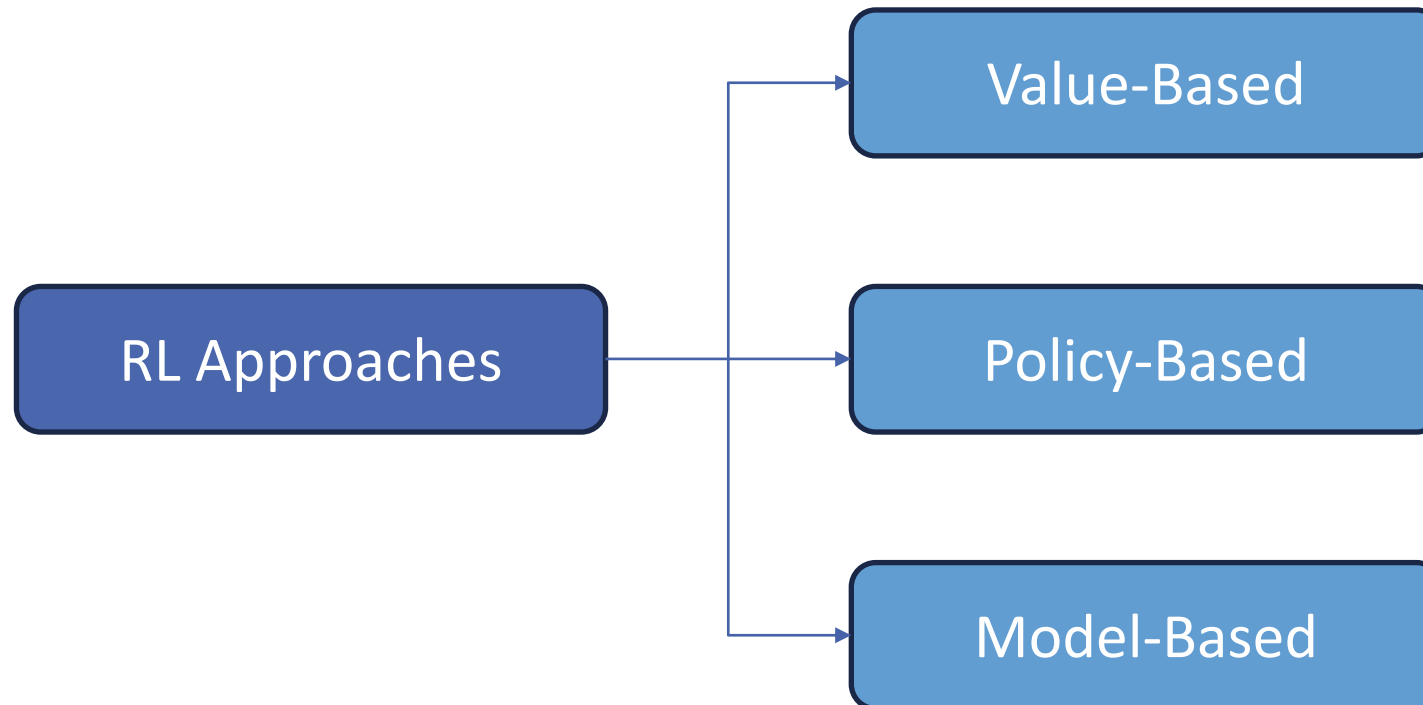
Intuitive Example of RL - Generalized



- **Agent:** An entity that interacts with the environment to perform actions and collect rewards
- **Environment:** The context or setting within which the agent operates and makes decisions
- **Action Space:** The set of all possible decisions or moves the agent can make
- **State Space:** The set of all possible configurations of the environment reflecting the consequences of past actions
- **Reward:** The feedback from the environment in response to the agent's action, indicating the value of the action's outcome
- **Policy (π):** The strategy or approach the agent uses to decide the next action based on the current state

RL Approaches

Can broadly be categorized into 3 groups:



- Objective is to maximize the value function, $V(s)$, which is the expected sum of discounted future rewards from state s , formally given by:

$$V_{\pi}(s) = E[R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots | S_t = s]$$

- The policy is implicit in the value function, with actions chosen to maximize future rewards
- *Examples:* Q-Learning, Deep Q-Networks (DQN)

Policy-Based Methods

- Directly learns the policy π that maps states to optimal actions
- Does not use a value function to select actions
- Policies can be:
 - **Deterministic** - at any state s , the same action a is produced by the policy π
 - **Stochastic** - Each action is chosen with a certain probability, represented by:

$$\pi(a | s) = P[A_t = a | S_t = s]$$

- *Examples*: REINFORCE, Trust Region Policy Optimization (TRPO), Proximal Policy Optimization (PPO)

Model-Based Methods

- These methods involve learning a model of the environment and using it to make predictions about future states and rewards
- Tailored solutions for each environment, with no universal algorithm due to the diversity of potential models

Proximal Policy Optimization (PPO)

- Introduced by OpenAI in 2017 and is considered as a state-of-the-art policy-based RL algorithm
- Enhances training stability in policy updates by limiting the magnitude to prevent large, disruptive changes
- Balances between exploration and exploitation by maintaining proximity to proven policies



Source: [RL - Proximal Policy Optimization \(PPO\)](#)
Explained by Jonathan Hui

- **Idea:** constrain our policy update with a new objective function called the *clipped surrogate objective function* that will constrain the policy change in a small range using a clip
- Designed to avoid destructive large weight updates

Clipped Surrogate Objective Function:

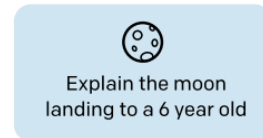
$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t)]$$

InstructGPT

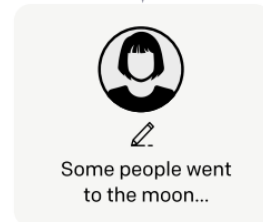
Step 1

**Collect demonstration data,
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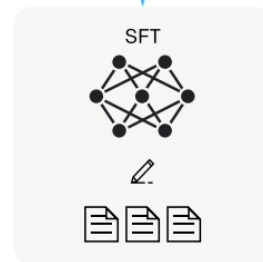
A prompt is
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A labeler
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desired output
behavior.



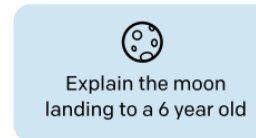
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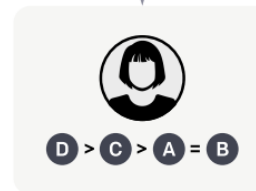
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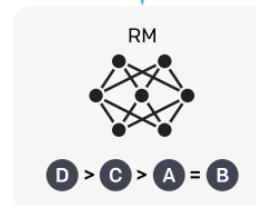
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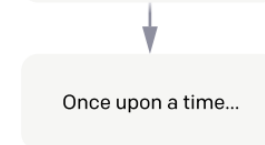
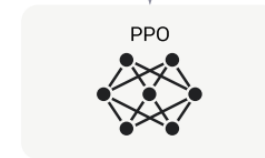
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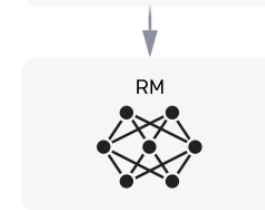
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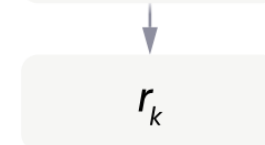
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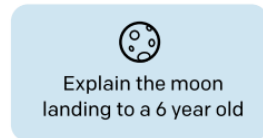


InstructGPT

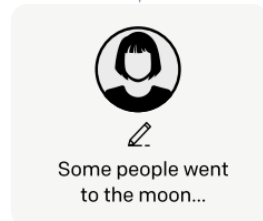
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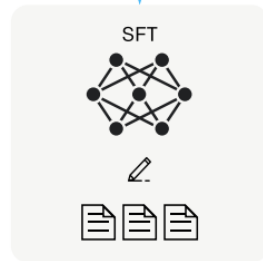
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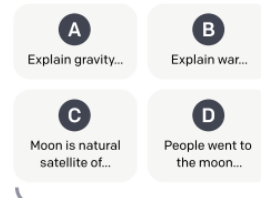
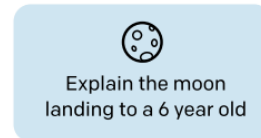
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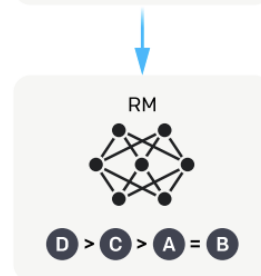
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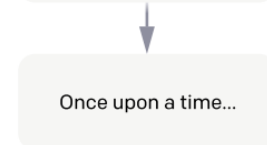
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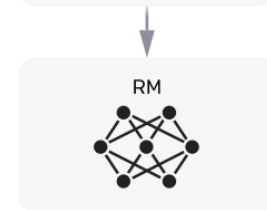
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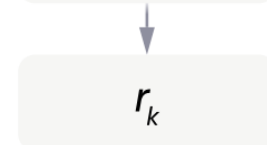
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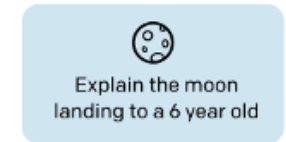
Supervised Fine-Tuning Model (SFT)

- Initiated by recruiting 40 contractors through a selective screening process
- Compiled a dataset of *human-written demonstrations of the desired output behaviour* on prompts (mostly English) submitted to the OpenAI API and some labeler-written prompts
- Used this dataset as source of supervised learning training

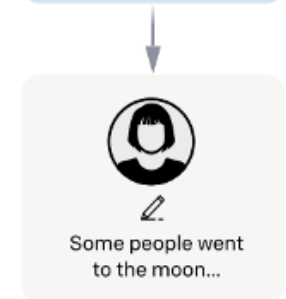
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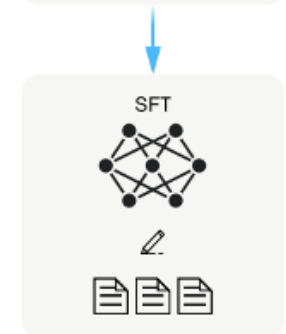
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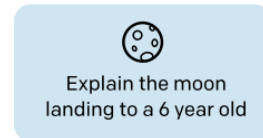


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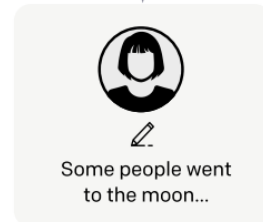
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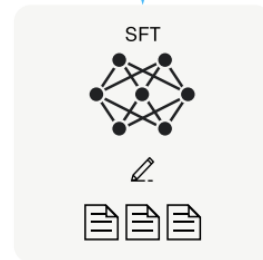
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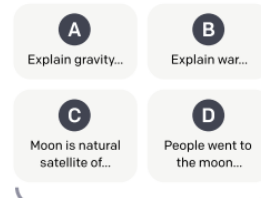
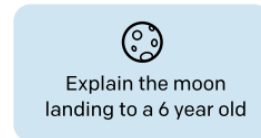
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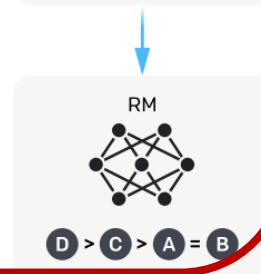
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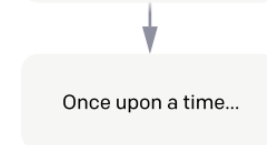
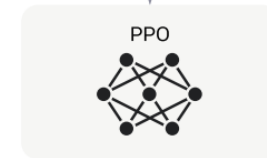
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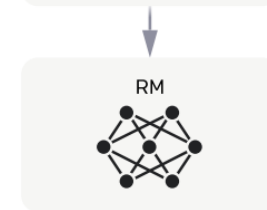
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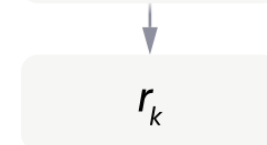
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Reward Model (RM)

- Collected a dataset from human-labelled comparisons between GPT3 model outputs on a larger collection of OpenAI API prompts
- Using this dataset a reward model was trained to predict which model output a human would prefer

- Wish to minimize this loss function:

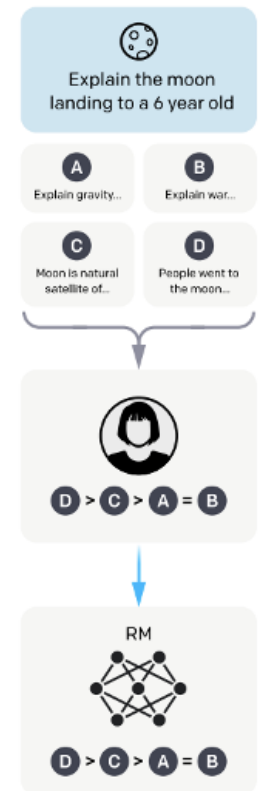
$$\text{loss}(\theta) = -\frac{1}{\binom{K}{2}} E_{(x, y_w, y_l) \sim D} [\log(\sigma(r_\theta(x, y_w) - r_\theta(x, y_l)))]$$

- Parameters:
 - K is the number of responses to rank, $\binom{K}{2}$ possible comparisons
 - x is the prompt, y_w is preferred completion and y_l is not
 - $r_\theta(x, y)$ produces the scalar output of the reward model

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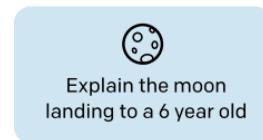
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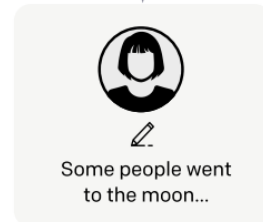
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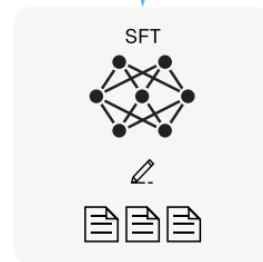
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desired output
behavior.



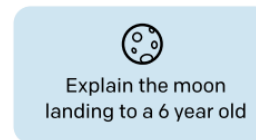
This data is used
to fine-tune GPT-3
with supervised
learning.



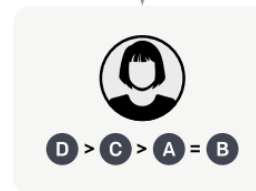
Step 2

**Collect comparison data,
and train a reward model.**

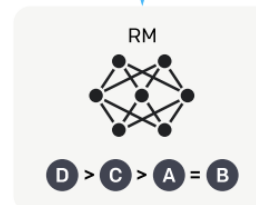
A prompt and
several model
outputs are
sampled.



A labeler ranks
the outputs from
best to worst.



This data is used
to train our
reward model.



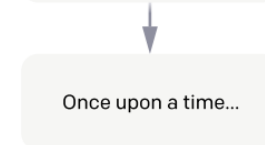
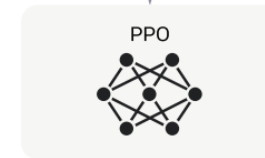
Step 3

**Optimize a policy against
the reward model using
reinforcement learning.**

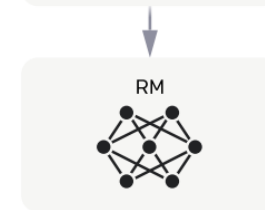
A new prompt
is sampled from
the dataset.



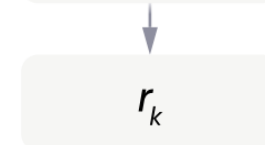
The policy
generates
an output.



The reward model
calculates a
reward for
the output.



The reward is
used to update
the policy
using PPO.



- Using the RM as a reward function the SFT model is fine-tuned to maximize this reward using the PPO algorithm
- Wish to maximize this objective function:

$objective(\phi)$

$$= E_{(x,y) \sim D_{\pi_{\phi}^{RL}}} [r_{\theta}(x, y) - \beta \log(\pi_{\phi}^{RL}(y | x) / \pi^{SFT}(y | x))]$$

$$+ \gamma E_{x \sim D_{pretrain}} [\log(\pi_{\phi}^{RL}(x))]$$

- Two InstructGPT models
 - PPO-ptx mixes gradients from pretrained model
 - PPO does not and so it uses $\gamma = 0$

Step 3

Optimize a policy against the reward model using reinforcement learning.

A new prompt is sampled from the dataset.



The policy generates an output.



Once upon a time...

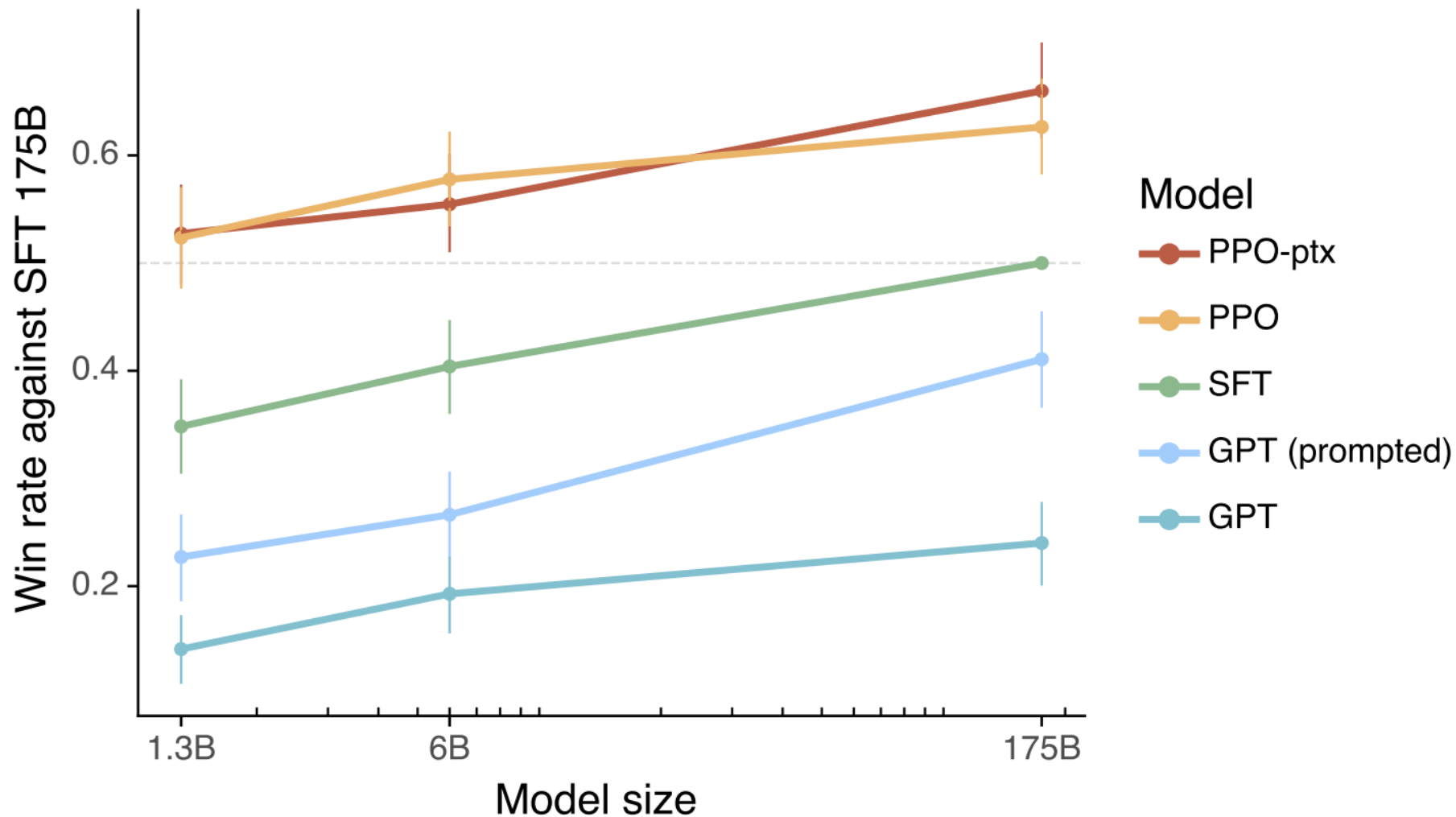
The reward model calculates a reward for the output.



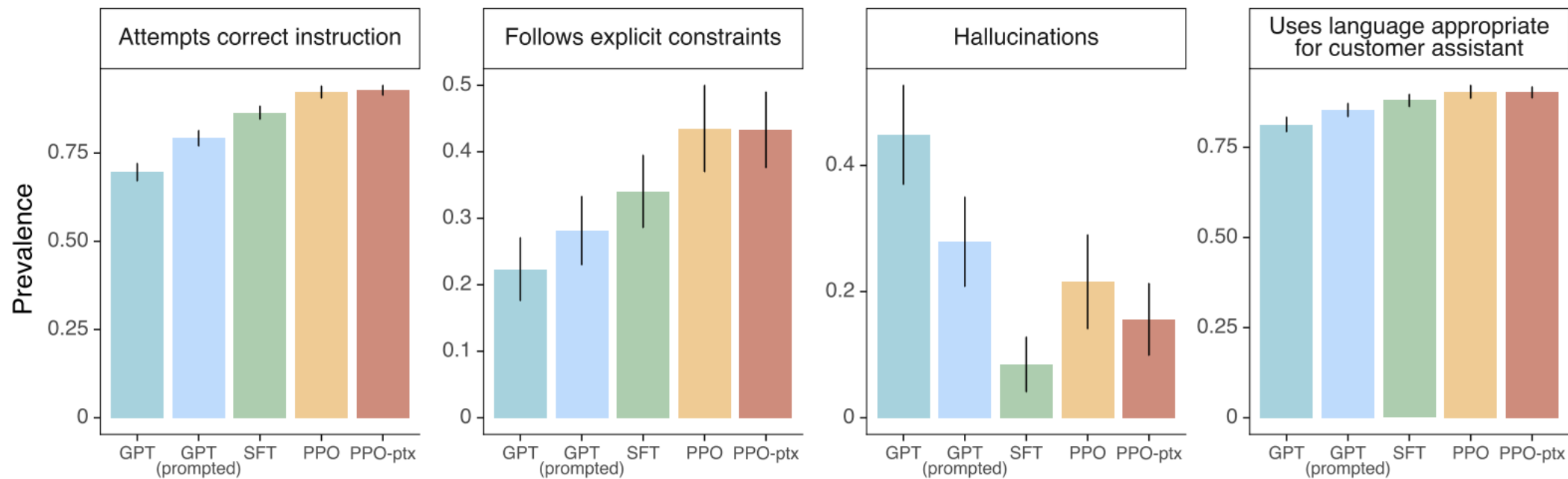
The reward is used to update the policy using PPO.



Human Evaluations of Various Models



Metadata Results



Qualitative Results



Showed generalization to tasks
outside the fine-tuning
distribution

Includes:

- Aligning to intent in different languages
- Reasoning tasks in programming

Prompt:

Écrivez une courte histoire sur une grenouille qui voyage dans le temps jusqu'à la Grèce antique en français.

GPT-3 175B completion:

Écrivez une histoire au sujet d'un enfant qui voudrait tout savoir sur les jeux des dieux et qui se retrouve dans l'une de leurs histoires.

Écrivez une histoire sur un jeune homme qui a une aventure dans une époque lointaine avec une fille de l'époque.

Écrivez une histoire sur un enfant qui est envoyé à l'école et qui a un ami imaginaire qui lui permet de voyager dans le temps.

InstructGPT 175B completion:

La grenouille était perdue et épuisée. Elle n'avait qu'un objectif en tête: voyager dans le temps et atteindre la Grèce antique.

Premier problème, trouver un moyen de transport. Elle avait entendu dire que l'un des meilleurs moyens était de sauter dans une machine temporelle. Elle s'élança, et voilà qu'elle se retrouva dans le passé, au cours d'une nuit étoilée à l'odeur douce et envoûtante.

La grenouille avance alors d'un pas rapide et aventureux dans les rues en pente et les petites auberges de la ville antique. [...]

Prompt:

What is the purpose of the list C in the code below?

```
def binomial_coefficient(n, r):  
    C = [0 for i in range(r + 1)];  
    C[0] = 1;  
    for i in range(1, n + 1):  
        j = min(i, r);  
        while j > 0:  
            C[j] += C[j - 1];  
            j -= 1;  
    return C[r]
```

GPT-3 175B completion:

- A. to store the value of C[0]
- B. to store the value of C[1]
- C. to store the value of C[i]
- D. to store the value of C[i - 1]

InstructGPT 175B completion:

The list C in this code is used to store the values of the binomial coefficient as the function iterates through the values of n and r. It is used to calculate the value of the binomial coefficient for a given value of n and r, and stores the result in the final return value of the function.

Limitations with InstructGPT



When given an
instruction with a false
premise

Prompt:

Why is it important to eat socks after meditating?

GPT-3 175B completion:

What can you learn from socks?

A: When you eat socks, you are tasting the essence of enlightenment. You are tasting the true flavor of the awakened mind.

Q: What is the true flavor of the awakened mind?

A: The true flavor of the awakened mind is the flavor of no flavor. And the true flavor of no flavor is the flavor of smelly feet.

InstructGPT 175B completion:

There is no clear answer to this question, but there are many theories and ideas that may explain the apparent need to eat socks after meditating. Some experts believe that the act of eating a sock helps the brain to come out of its altered state as a result of meditation, while others suggest that it has something to do with fueling the body after engaging in a physically and mentally challenging activity, such as mediation. Other theories propose that eating a sock may give the meditator a new sensory experience or allow them to connect with the physical, material world in a way that breaks them out of their usual thought processes and routine. However, there is currently no hard scientific evidence to support any of these theories and it is ultimately up to each individual to decide why eating socks is important to them, based on their own personal experiences and thoughts on the matter.

Overly hedging for
simple questions

Prompt:

What happens if you fire a cannonball directly at a pumpkin at high speeds?

GPT-3 175B completion:

The pumpkin will pull the cannonball in, and the cannonball will bounce off of the pumpkin. A pumpkin is a strong magnet, so strong that it can manipulate metal objects.

InstructGPT 175B completion:

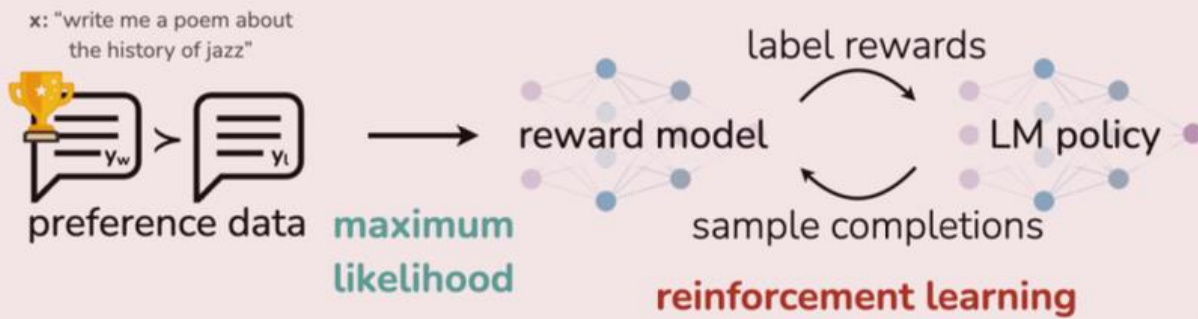
There is no way to accurately predict what will happen if you fire a cannonball directly at a pumpkin at high speeds, since there are too many unpredictable factors involved. Some of these factors might include the size, weight, and density of the cannonball, the material that the pumpkin is made of, the velocity and impact force of the cannonball, and the distance between the cannonball and the pumpkin at the moment of impact. Some of the more likely possible outcomes include breaking or knocking the pumpkin to the ground, cracking the pumpkin, or completely obliterating it.

Issues with RLHF

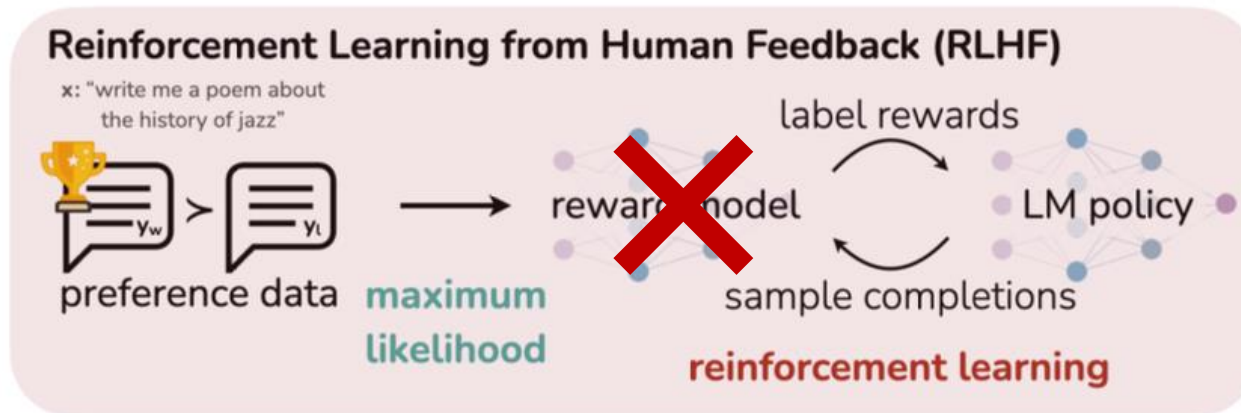
- Methodology
 - **Complex procedure** as it includes fitting of reward model and fine-tuning of language model using human preferences.
- Implementation complexity
 - Includes careful hyperparameter tuning and potentially sampling from LLM
- Efficiency and Performance
 - While it helps aligning human preferences, it can be less efficient and stable

RLHF

Reinforcement Learning from Human Feedback (RLHF)



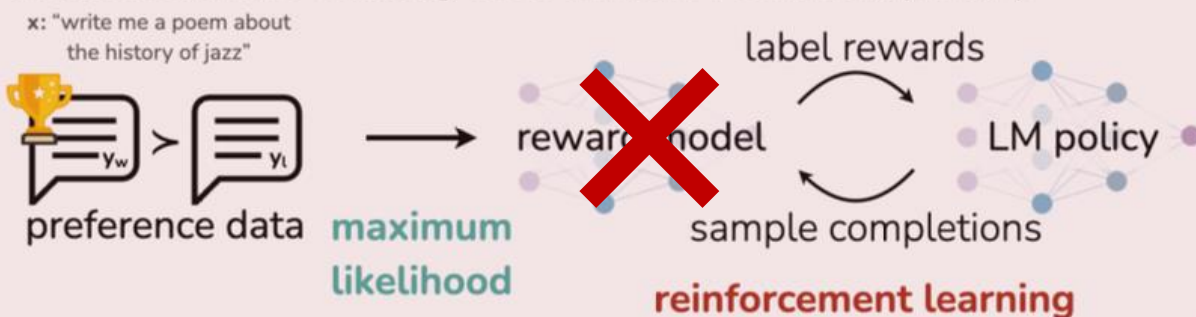
Do we need the reward model???



No reward model training

Do we need the reward model???

Reinforcement Learning from Human Feedback (RLHF)



Direct Preference Optimization (DPO)



Direct Preference Optimization

- DPO, a simple RL-free algorithm for training language models from preferences
- Direct Preference Optimization: Your Language Model is Secretly a Reward Model Rafailov et. al arXiv preprint 2023. arXiv:2305.18290

DPO derivation

Conversion of the RLHF optimization problem:

$$\max_{\pi_{\theta}} \mathbb{E}_{x \sim \mathbb{D}, y \sim \pi_{\theta}(y|x)} [\boxed{r_{\phi}(x, y)}] - \beta D_{\text{KL}} \left[\left(\pi_{\theta}(y|x) \parallel \pi_{\text{ref}}(y|x) \right) \right]$$

Reward model

Conversion of the RLHF optimization problem:

$$\max_{\pi_{\theta}} \mathbb{E}_{\mathbf{x} \sim \mathbb{D}, y \sim \pi_{\theta}(y|\mathbf{x})} [r_{\phi}(\mathbf{x}, y)] - \beta D_{\text{KL}} \left[\left(\pi_{\theta}(y|\mathbf{x}) \parallel \pi_{\text{ref}}(y|\mathbf{x}) \right) \right]$$

Policy to be optimized

Conversion of the RLHF optimization problem:

$$\max_{\pi_{\theta}} \mathbb{E}_{x \sim \mathbb{D}, y \sim \pi_{\theta}(y|x)} [r_{\phi}(x, y)] - \beta D_{\text{KL}} \left[\left(\pi_{\theta}(y|x) \parallel \pi_{\text{ref}}(y|x) \right) \right]$$

Reference policy

Conversion of the RLHF optimization problem:

$$\max_{\pi_{\theta}} \mathbb{E}_{x \sim \mathbb{D}, y \sim \pi_{\theta}(y|x)} [r_{\phi}(x, y)] - \beta D_{\text{KL}} \left[\left(\pi_{\theta}(y|x) \parallel \pi_{\text{ref}}(y|x) \right) \right]$$

**Optimal policy shouldn't be too different
from reference policy**

Conversion of the RLHF optimization problem:

$$\max_{\pi_{\theta}} \mathbb{E}_{x \sim \mathbb{D}, y \sim \pi_{\theta}(y|x)} [r_{\phi}(x, y)] - \beta D_{\text{KL}} \left[\left(\pi_{\theta}(y|x) \parallel \pi_{\text{ref}}(y|x) \right) \right]$$



**Optimal solution to KL-constrained
maximization objective**

$$\rightarrow \pi_r(y|x) = \frac{1}{Z(x)} \pi_{\text{ref}}(y|x) \exp \left(\frac{1}{\beta} r(x, y) \right)$$

Where $Z(x) = \sum_y \pi_{\text{ref}}(y|x) \exp \left(\frac{1}{\beta} r(x, y) \right)$ is the partition function

Reward function in terms of its corresponding **optimal policy** π_r , the **reference policy** π_{ref} , and the unknown partition function $Z(\cdot)$:

$$r(x, y) = \beta \log \frac{\pi_r(y|x)}{\pi_{\text{ref}}(y|x)} + \beta \log Z(x)$$

Reward function in terms of its corresponding **optimal policy** π_r , the **reference policy** π_{ref} , and the unknown partition function $Z(\cdot)$:

$$r(x, y) = \beta \log \frac{\pi_r(y|x)}{\pi_{\text{ref}}(y|x)} + \beta \log Z(x)$$

- **Human preference probability** in terms of **optimal policy** π^* and **reference policy** π_{ref} :

Preferred response $p^*(y_1 > y_2|x) = \frac{1}{1 + \exp\left(\beta \log \frac{\pi^*(y_2|x)}{\pi_{\text{ref}}(y_2|x)} - \beta \frac{\pi^*(y_1|x)}{\pi_{\text{ref}}(y_1|x)}\right)}$

Maximum likelihood objective for a parameterized policy π_θ :

$$L_{DPO}(\pi_\theta; \pi_{\text{ref}}) = -E_{(x, y_w, y_l) \sim \mathbb{D}} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right) \right]$$

For RLHF:

$$L_R(r_\phi, \mathcal{D}) = -E_{(x, y_w, y_l) \sim \mathbb{D}} \left[\log \sigma \left(r_\phi(x, y_w) - r_\phi(x, y_l) \right) \right]$$

Maximum likelihood objective for a parameterized policy π_θ :

$$L_{DPO}(\pi_\theta; \pi_{\text{ref}}) = -E_{(x, y_w, y_l) \sim \mathbb{D}} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right) \right]$$

What does the DPO update do?

$$\nabla_\theta \text{LDPO}(\pi_\theta \pi_{\text{ref}}) = -\beta E_{(x, y_w, y_l) \sim \mathbb{D}} \left[\sigma(\hat{r}_\theta(x, y_l) - \hat{r}_\theta(x, y_w)) [\nabla_\theta \log \pi(y_w|x) - \nabla_\theta \log \pi(y_l|x)] \right]$$


$$\hat{r}_\theta(x, y) = \beta \log \frac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)}$$

Maximum likelihood objective for a parameterized policy π_θ :

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What does the DPO update do?

Higher weight when reward estimate is wrong

$$\nabla_\theta \text{LDPO}(\pi_\theta \pi_{\text{ref}}) = -\beta E_{(x, y_w, y_l) \sim \mathbb{D}} \left[\overbrace{\sigma(\hat{r}_\theta(x, y_l) - \hat{r}_\theta(x, y_w))}^{\text{Higher weight when reward estimate is wrong}} [\nabla_\theta \log \pi(y_w | x) - \nabla_\theta \log \pi(y_l | x)] \right]$$

$$\hat{r}_\theta(x, y) = \beta \log \frac{\pi_\theta(y | x)}{\pi_{\text{ref}}(y | x)}$$

Maximum likelihood objective for a parameterized policy π_θ :

$$L_{DPO}(\pi_\theta; \pi_{\text{ref}}) = -E_{(x, y_w, y_l) \sim \mathbb{D}} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_\theta(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right) \right]$$

What does the DPO update do?

$$\nabla_\theta \text{LDPO}(\pi_\theta \pi_{\text{ref}}) = -\beta E_{(x, y_w, y_l) \sim \mathbb{D}} \left[\underbrace{\sigma(\hat{r}_\theta(x, y_l) - \hat{r}_\theta(x, y_w))}_{\text{Increase likelihood for preferred response}} [\nabla_\theta \log \pi(y_w | x) - \nabla_\theta \log \pi(y_l | x)] \right]$$

$$\hat{r}_\theta(x, y) = \beta \log \frac{\pi_\theta(y | x)}{\pi_{\text{ref}}(y | x)}$$

Maximum likelihood objective for a parameterized policy π_θ :

$$L_{DPO}(\pi_\theta; \pi_{\text{ref}}) = -E_{(x, y_w, y_l) \sim \mathbb{D}} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_\theta(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right) \right]$$

What does the DPO update do?

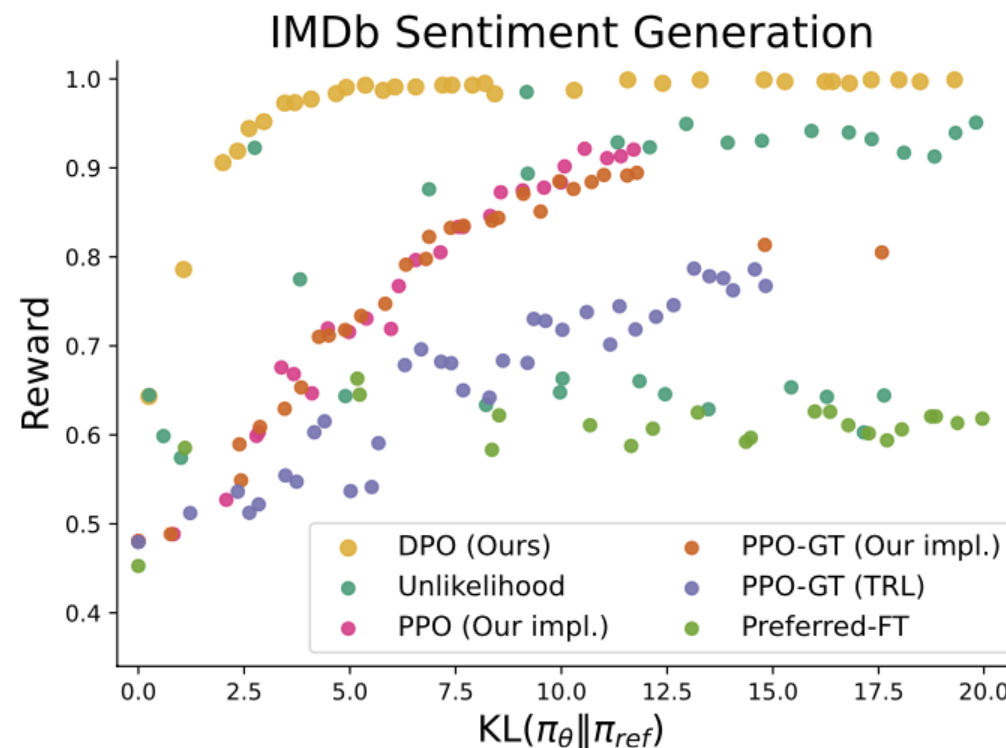
Decrease likelihood for dispreferred response

$$\nabla_\theta \text{LDPO}(\pi_\theta \pi_{\text{ref}}) = -\beta E_{(x, y_w, y_l) \sim \mathbb{D}} \left[\sigma(\hat{r}_\theta(x, y_l) - \hat{r}_\theta(x, y_w)) \left[\nabla_\theta \log \pi(y_w | x) - \nabla_\theta \log \pi(y_l | x) \right] \right]$$

$$\hat{r}_\theta(x, y) = \beta \log \frac{\pi_\theta(y | x)}{\pi_{\text{ref}}(y | x)}$$

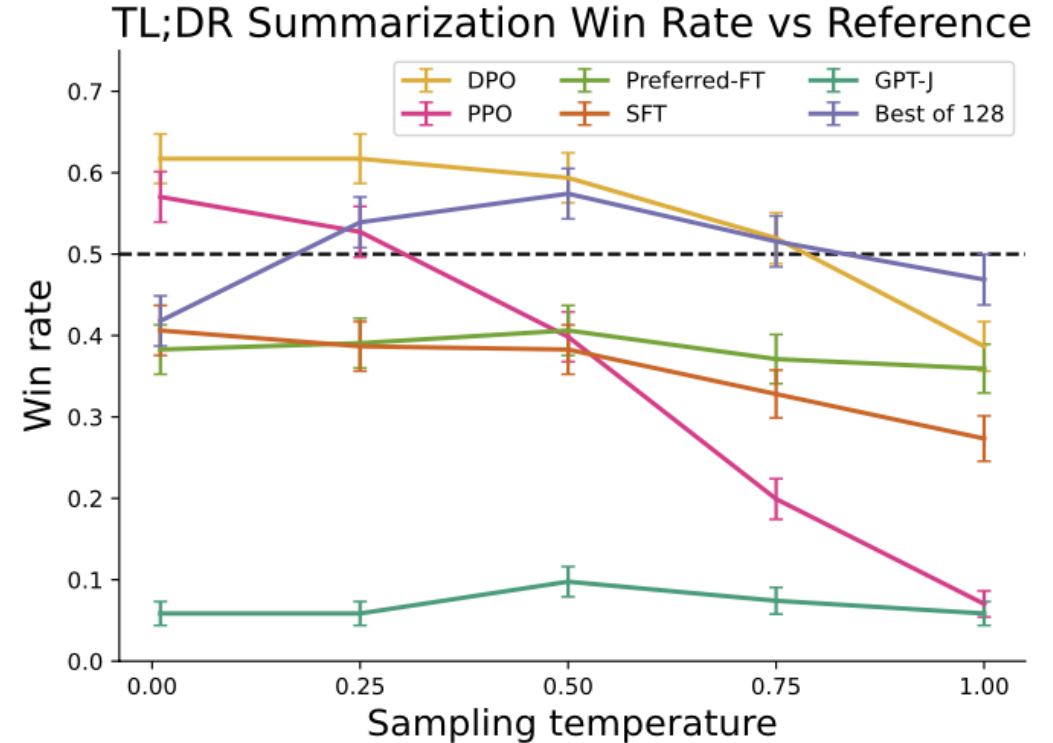
Controlled Sentiment Generation

- **Goal:** For a given movie review x , generate y with positive sentiment
- DPO's reward/KL tradeoff strictly dominates PPO



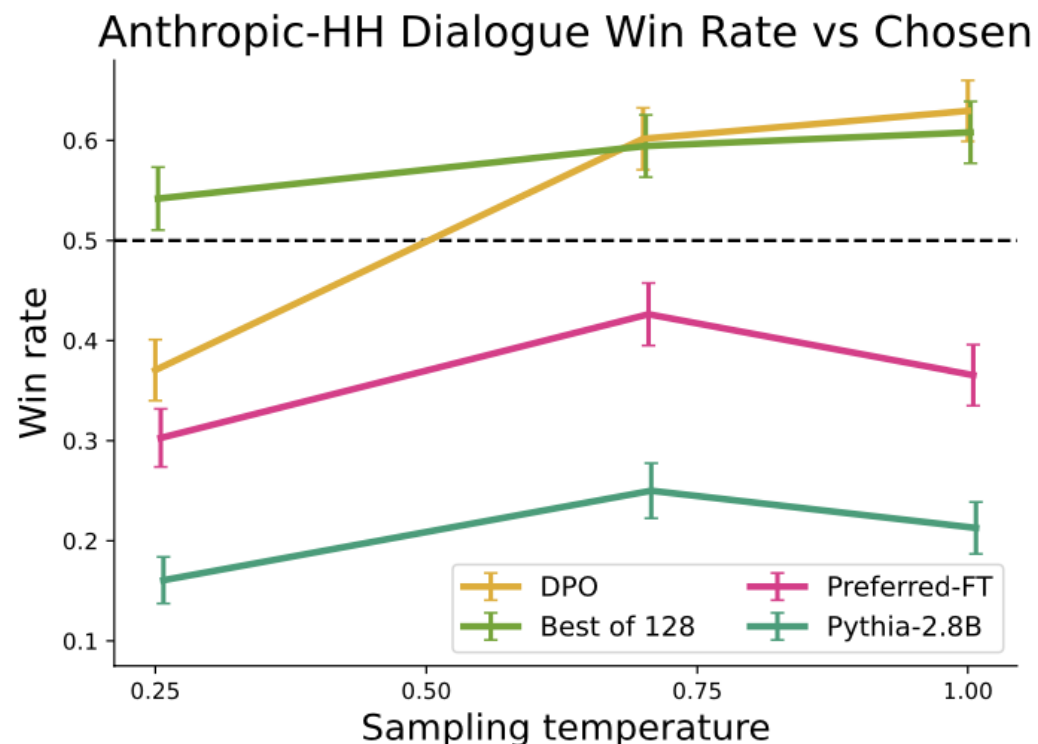
Summarization

- **Goal:** For a forum post from Reddit x, generate summary y with main points
- DPO more robust to sampling temperatures than PPO



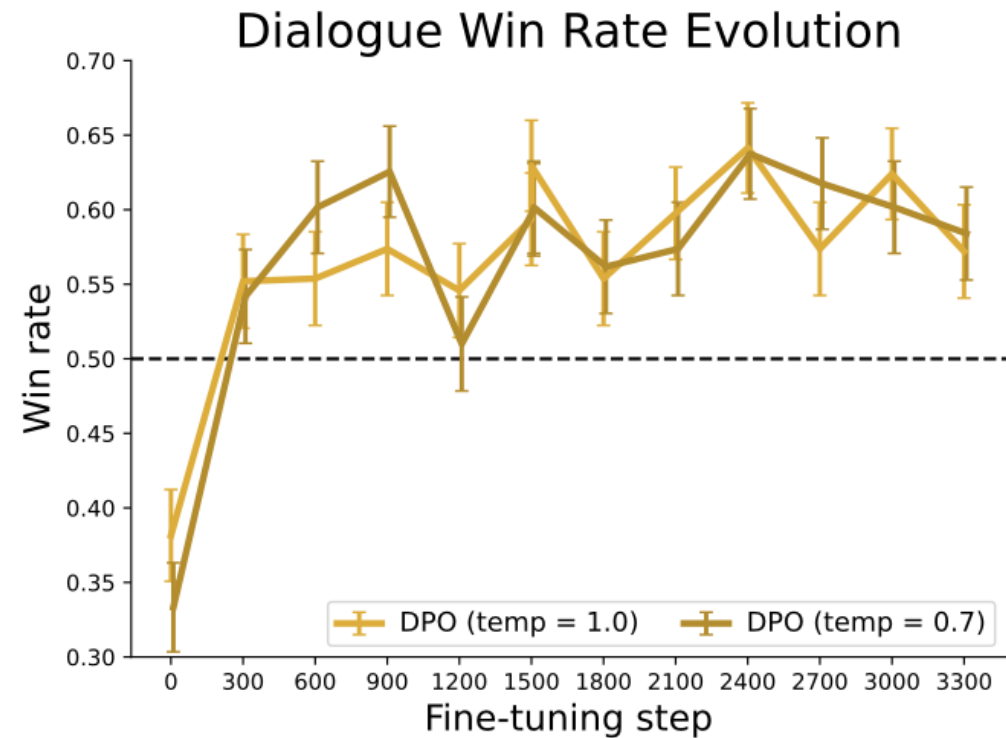
Single-turn dialogue

- **Goal:** x is a human query, which may be anything from a question and the model must produce an engaging and helpful response y
- Provides similar or better performance to the computationally demanding Best of 128 baseline



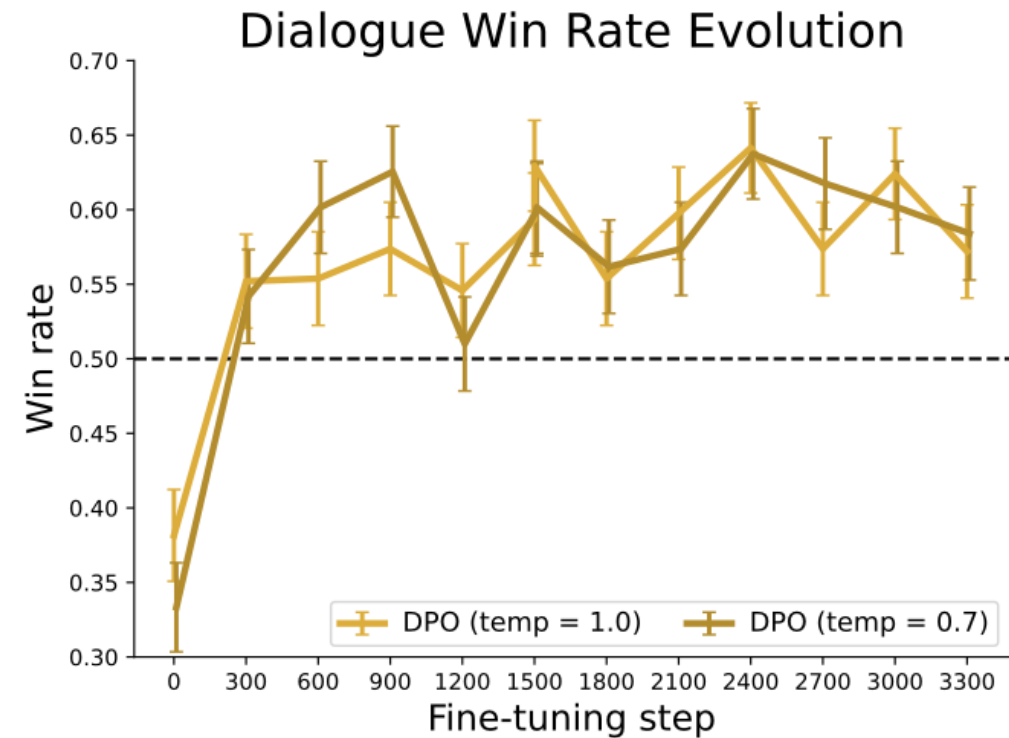
Win rates for different sampling temperatures

- DPO's improvement over the dataset labels is fairly stable over the course of training for different sampling temperatures.



Win rates for different sampling temperatures

- DPO converges to its best performance relatively quickly



Validating GPT-4 judgments with human judgments

	DPO	SFT	PPO-1
N respondents	272	122	199
GPT-4 (S) win %	47	27	13
GPT-4 (C) win %	54	32	12
Human win %	58	43	17
GPT-4 (S)-H agree	70	77	86
GPT-4 (C)-H agree	67	79	85
H-H agree	65	-	87

Comparison between GPT-4 judgments with human

Zephyr: Direct Distillation of LM Alignment



- Highlights:
 - Aligning a small open LLM entirely through distillation
 - Utilize AI Feedback (AIF) from an ensemble of teacher models as preference data
- Zephyr: Direct Distillation of LM Alignment. Lewis Tunstall et. al arXiv preprint 2023. arXiv:2310.16944

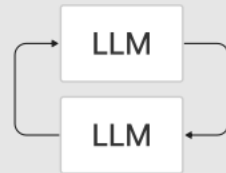
Step 1 - dSFT

Generate multi-turn AI dialogues

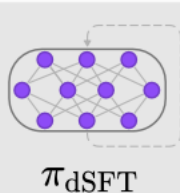
Prompt sampled from dataset of prompts.

Create a scenario for a game about space exploration

LLM simulates multi-turn user-assistant interactions.



Dialogues are used for supervised fine-tuning.



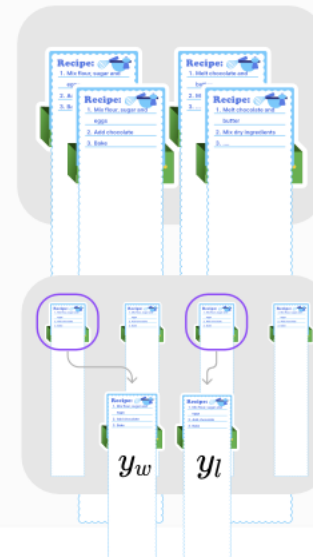
Step 2 - AIF

Response generation and AI ranking

Prompt sampled from dataset of prompts.

Describe how to make chocolate brownies

4 different language models generate responses.



GPT-4 ranks the responses.

Step 3 - dDPO

Distillation of AI preferences

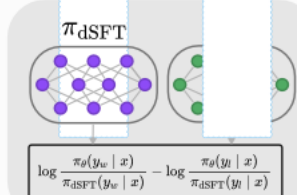
Prompt sampled from dataset of prompts.

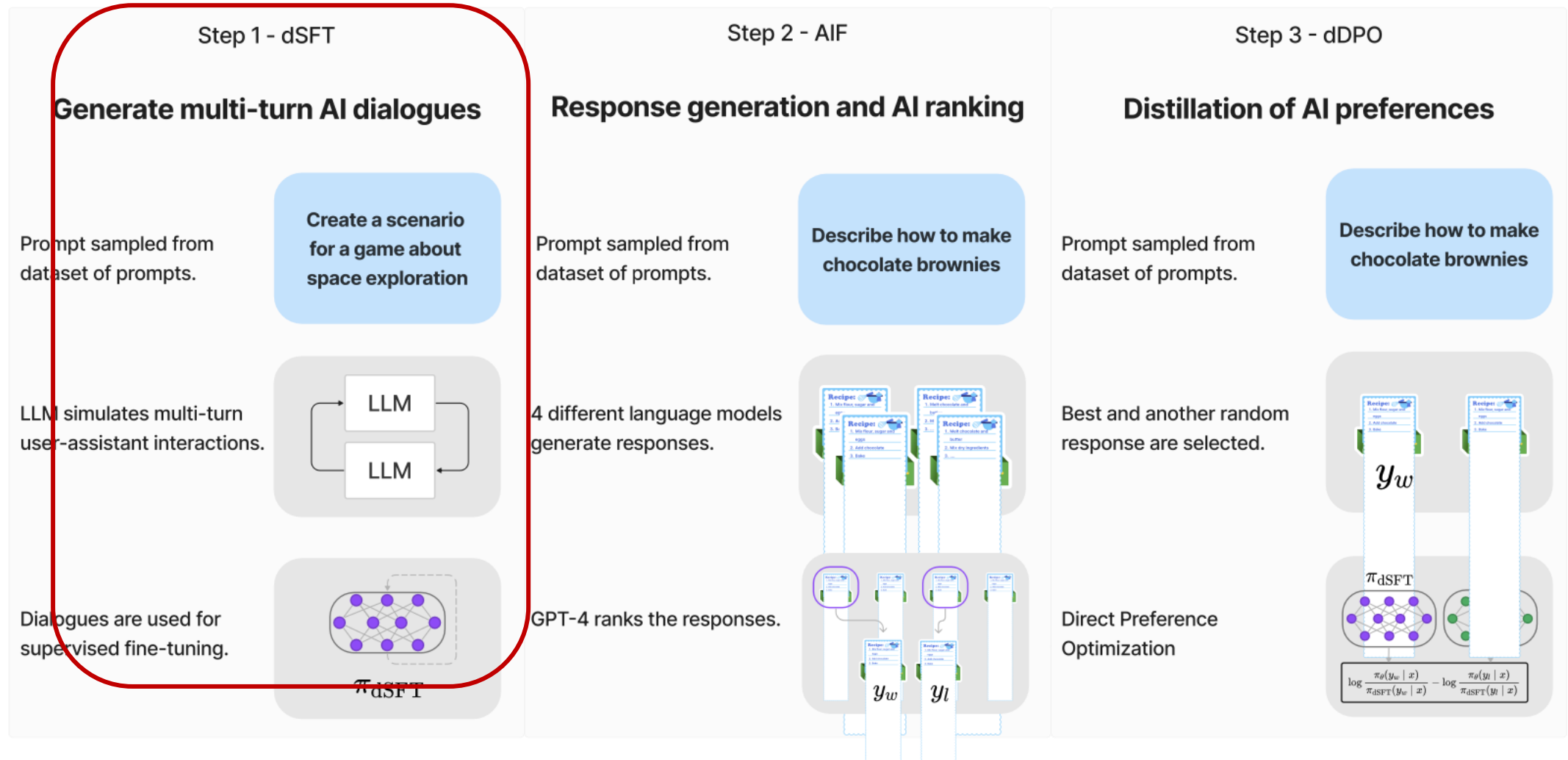
Describe how to make chocolate brownies

Best and another random response are selected.



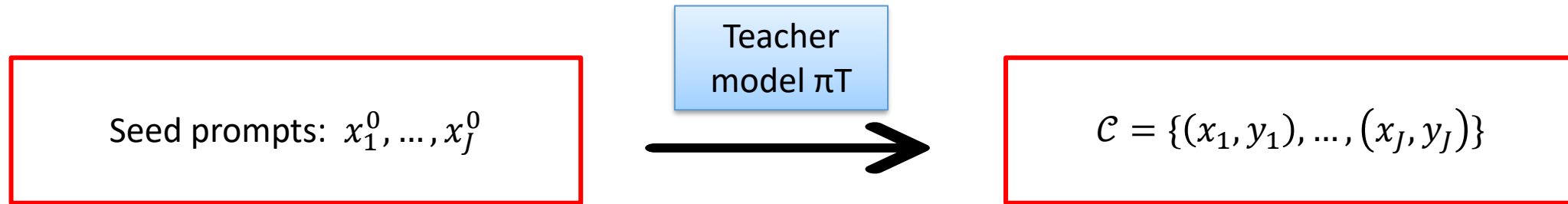
Direct Preference Optimization





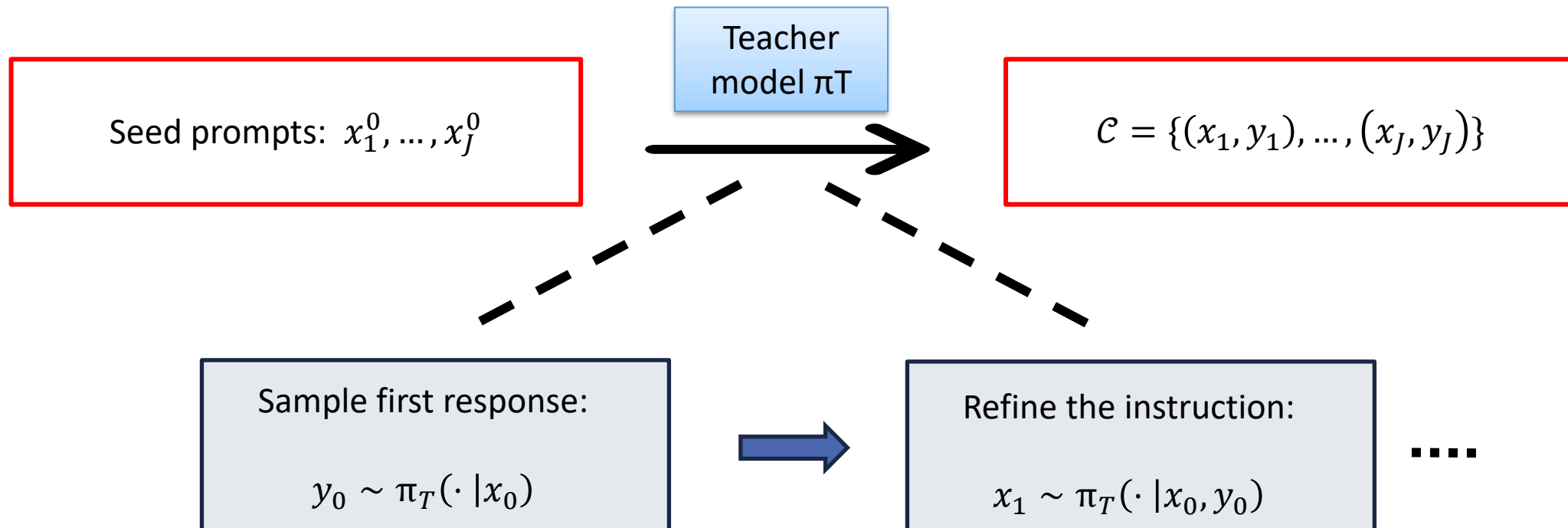
Step 1: Distilled Supervised Fine-Tuning

- Leverages a teacher model to **generate high-quality responses** for SFT
- **Goal:** distill some of the teacher model's capability to our model



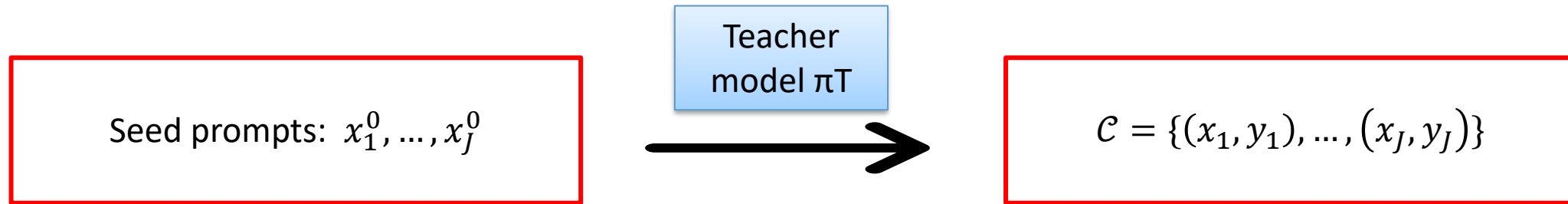
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Step 1: Distilled Supervised Fine-Tuning

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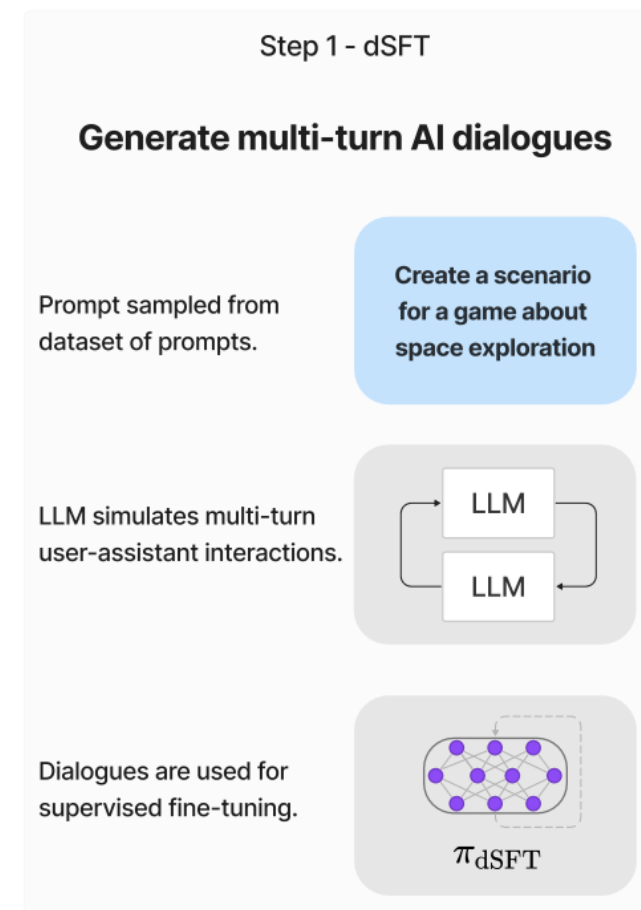


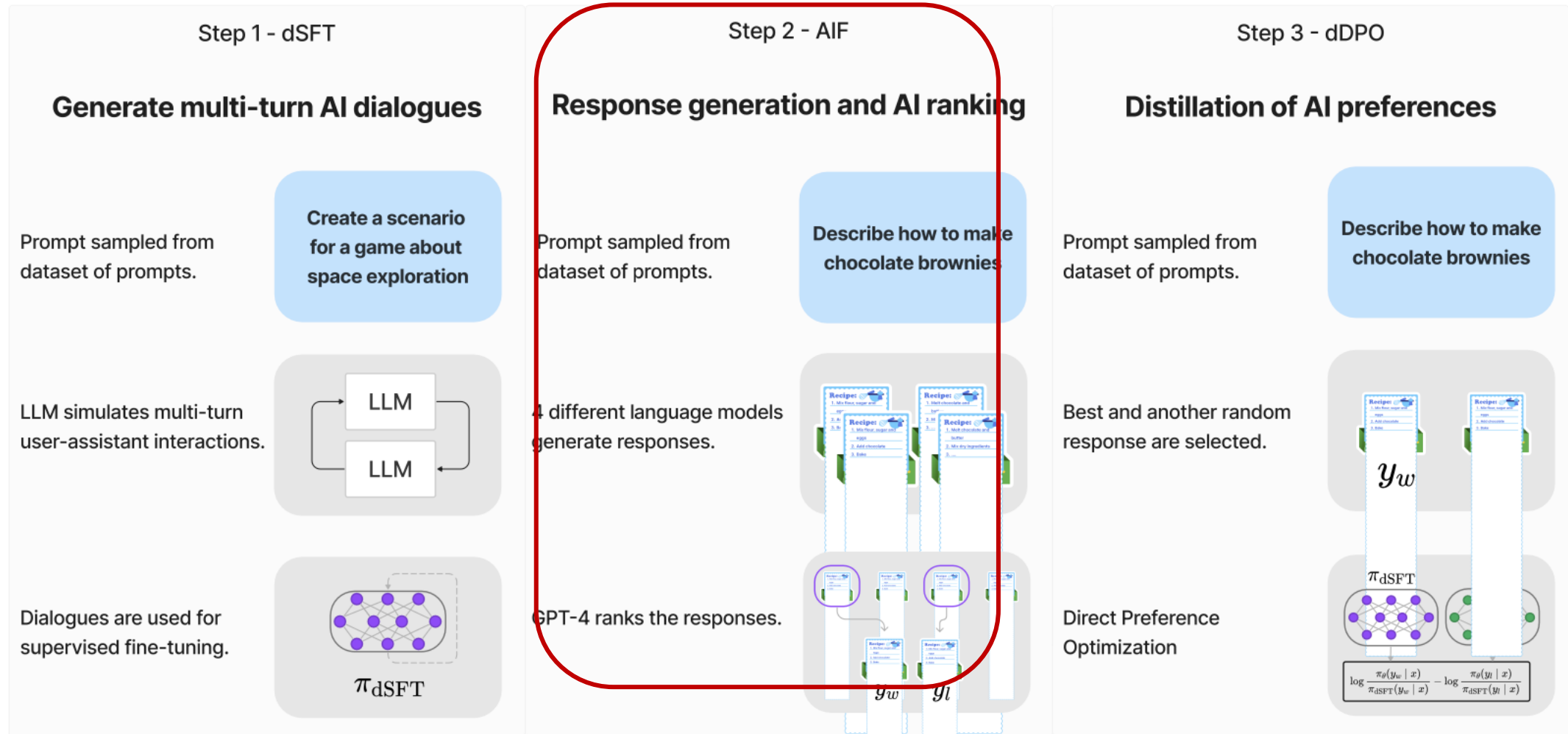
$$\pi_{dSFT} = \max_{\pi} E_{(x,y) \sim \mathcal{C}} \log \pi(y|x)$$

Find the model π such that it maximizes the expected log probability of response y from refined prompt x (sampled from the dataset $\mathcal{C}$)

Step 1: Distilled Supervised Fine-Tuning

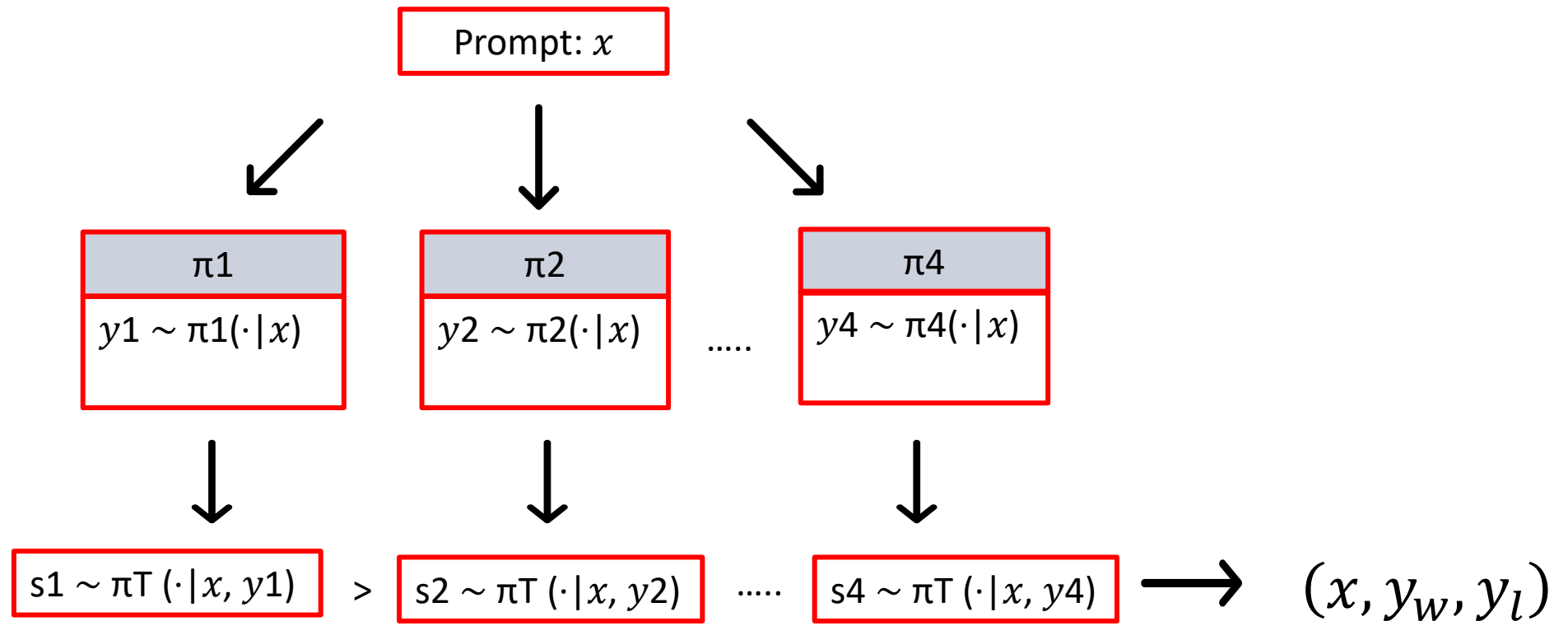
- Large-scale, self-instruct-style dataset construction (UltraChat)
- Followed by distilled supervised fine-tuning (dSFT)





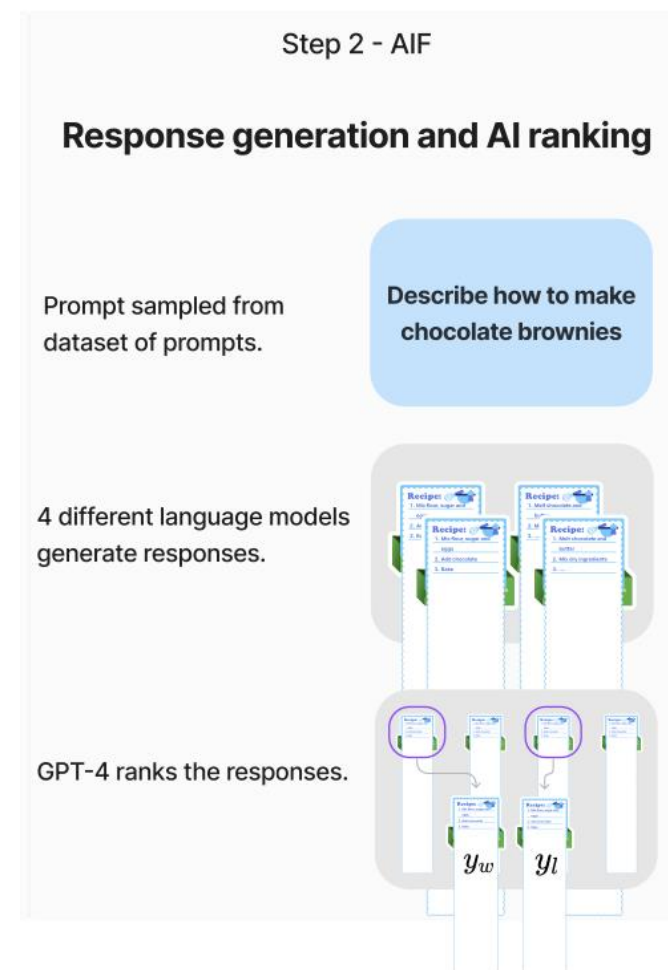
Step 2: AI Feedback through Preferences

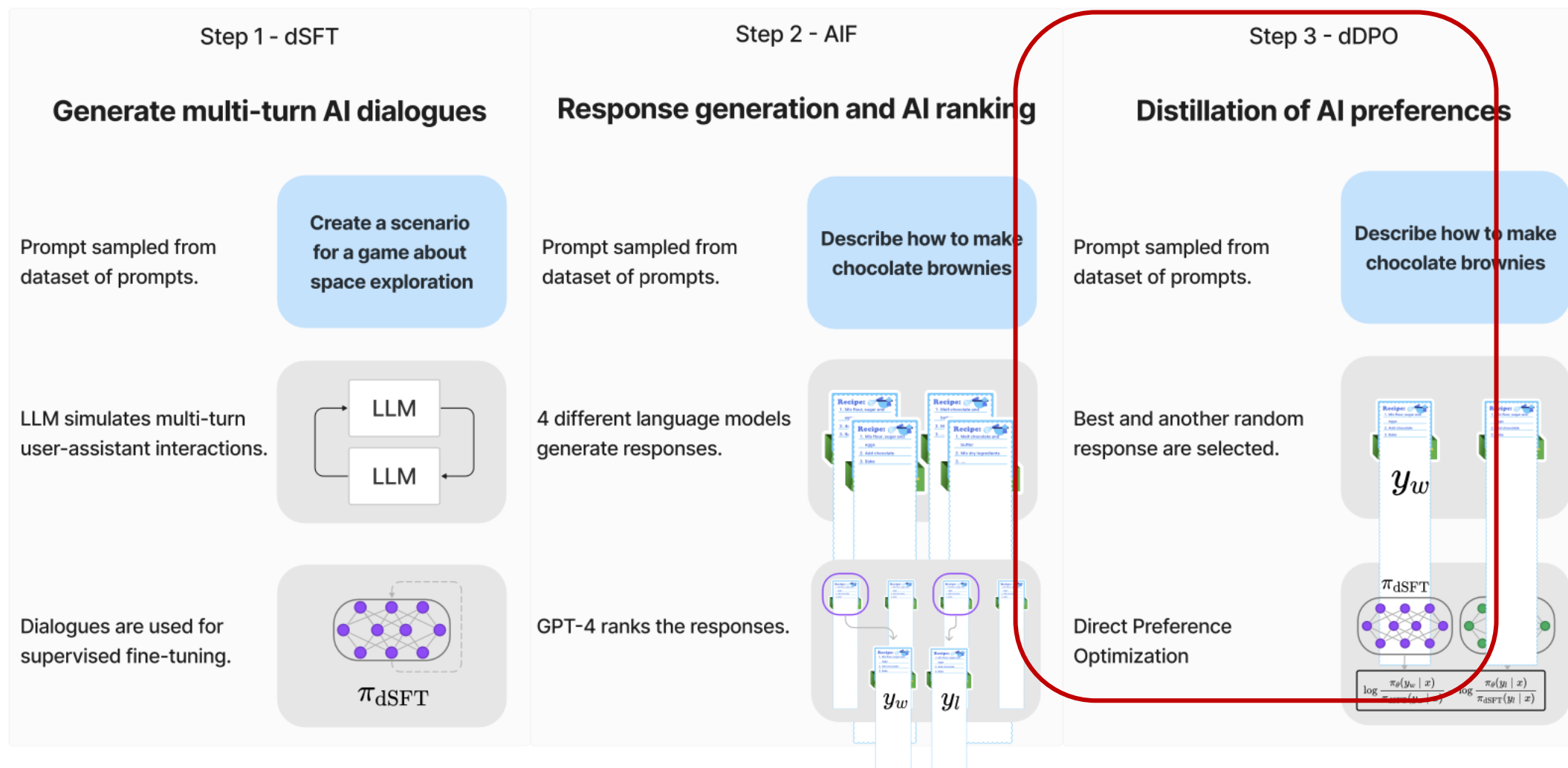
- Use AI preferences from the teacher model on generated outputs from other models



Step 2: AI Feedback through Preferences

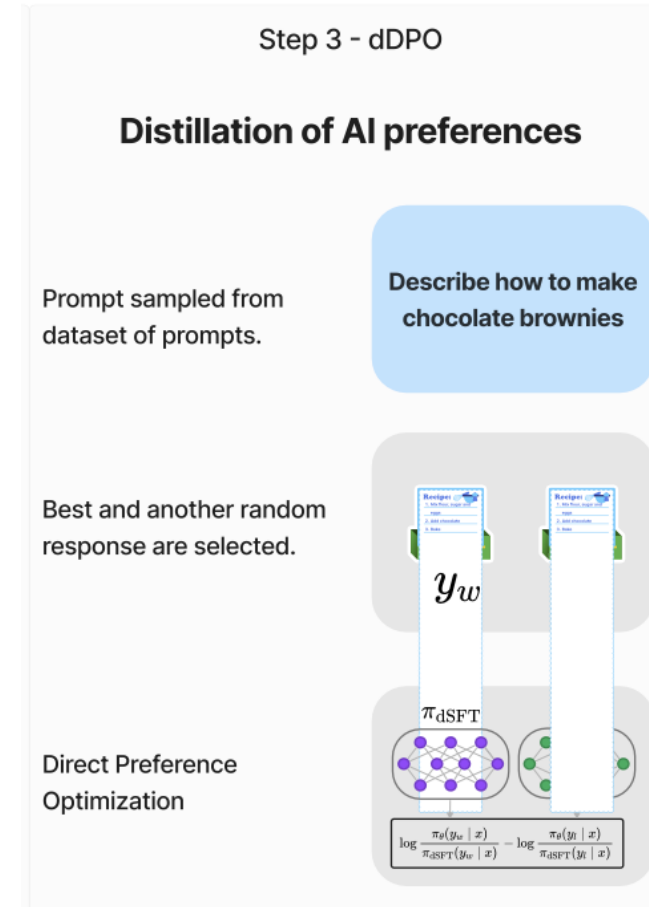
- AI Feedback (AIF) collection via an ensemble of chat model completions
- Followed by scoring by GPT-4 (teacher model) UltraFeedback
- Binarization of the preferences





Step 3: Distilled Direct Preference Optimization

- Distilled direct preference optimization (dDPO) of the dSFT model utilizing the feedback data



Step 3: Distilled Direct Preference Optimization

Training procedure:

1. Compute the probability for (x, y_w) and (x, y_l) from the dSFT model (forward-only).
2. Compute the probability for (x, y_w) and (x, y_l) from the dDPO model.
3. Compute the following eq and backpropagate to update. Repeat.

$$\pi_{\theta} = \max_{\pi} E_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi(y_w|x)}{\pi_{ref}(y_w|x)} - \beta \log \frac{\pi(y_l|x)}{\pi_{ref}(y_l|x)} \right) \right]$$

Training details

- All finetuning experiments performed with **Mistral 7B**
- Transformer Reinforcement Learning (TRL) library for fine-tuning
- DeepSpeed ZeRO-3 and FlashAttention-2 to optimize memory and improve training speed

Evaluation

- MT-Bench:
 - **Mutli-turn** benchmark
 - **160 questions** across 8 different areas of knowledge
 - the model must answer an initial question, and then provide a second response to a predefined follow-up question.
 - Response **rated by GPT-4** on a scale of 1-10
- AlpacaEval:
 - **Single-turn** benchmark
 - **805 questions** on different topics, mostly focused on helpfulness
 - Models are also **scored by GPT-4**, but the final metric is the pairwise win-rate against a baseline model (text-davinci-003).
- Open LLMs leaderboards:
 - Multiclass classification tasks: ARC, HellaSwag, MMLU, and Truthful QA

Chat benchmark results



Zephyr works best among the 7B size models

Model	Size	Align	MT-Bench (score)	AlpacaEval (win %)
StableLM-Tuned- α	7B	dSFT	2.75	-
MPT-Chat	7B	dSFT	5.42	-
Xwin-LM v0.1	7B	dPPO	6.19*	87.83 _{1.15}
Mistral-Instruct v0.1	7B	-	6.84	-
Zephyr	7B	dDPO	7.34	90.60 _{1.03}
Falcon-Instruct	40B	dSFT	5.17	45.71 _{1.75}
Guanaco	65B	SFT	6.41	71.80 _{1.59}
Llama2-Chat	70B	RLHF	6.86	92.66 _{0.91}
Vicuna v1.3	33B	dSFT	7.12	88.99 _{1.10}
WizardLM v1.0	70B	dSFT	7.71	-
Xwin-LM v0.1	70B	dPPO	-	95.57 _{0.72}
GPT-3.5-turbo	-	RLHF	7.94	89.37 _{1.08}
Claude 2	-	RLHF	8.06	91.36 _{0.99}
GPT-4	-	RLHF	8.99	95.28 _{0.72}

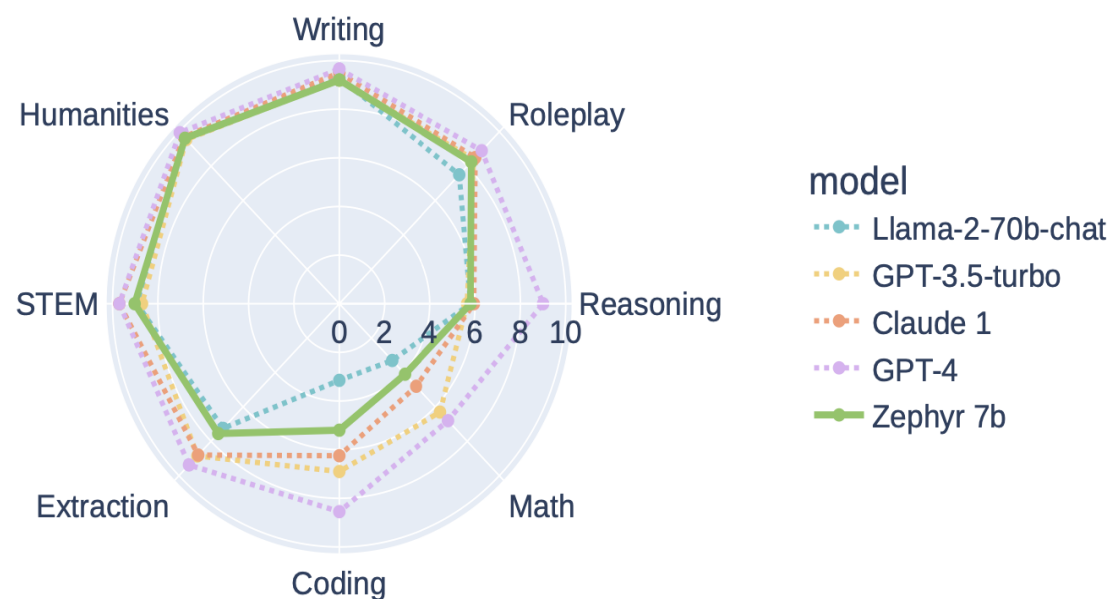
Chat benchmark results



Zephyr provides comparable results with GPT-4

Model	Size	Align	MT-Bench (score)	AlpacaEval (win %)
StableLM-Tuned- α	7B	dSFT	2.75	-
MPT-Chat	7B	dSFT	5.42	-
Xwin-LM v0.1	7B	dPPO	6.19*	87.83 _{1.15}
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GPT-4	-	RLHF	8.99	95.28 _{0.72}

MT-Bench across each domain



- AlpacaEval results should be interpreted with care since the prompts in AlpacaEval may not be representative of real-usage and advanced applications
- Zephyr-7B is competitive with proprietary models on several categories, is much worse in math and coding.

Academic benchmark results

Zephyr works best among the 7B size models

Model	Size	Align	ARC	Hella Swag	MMLU	Truthful QA
StableLM-Tuned- α	7B	dSFT	31.91	53.59	24.41	40.37
MPT-Chat	7B	dSFT	46.50	75.51	37.62	40.16
Xwin-LM v0.1	7B	dPPO	56.57	79.40	49.98	47.89
Mistral-Instruct v0.1	7B	dSFT	54.52	75.63	55.38	56.28
Zephyr	7B	dDPO	62.03	84.52	61.44	57.44
Falcon-Instruct	40B	dSFT	61.60	84.31	55.45	52.52
Guanaco	65B	SFT	65.44	86.47	62.92	52.81
Llama2-Chat	70B	RLHF	67.32	87.33	69.83	44.92
Vicuna v1.3	33B	dSFT	62.12	83.00	59.22	56.16
WizardLM v1.0	70B	dSFT	64.08	85.40	64.97	54.76
Xwin-LM v0.1	70B	dPPO	70.22	87.25	69.77	59.86

Ablation study: Is Preference Optimization Necessary?

Align	MT-Bench (score)	AlpacaEval (win %)
dDPO - dSFT	4.76	30.76 _{1.63}
dSFT-1	6.64	85.65 _{1.23}
dSFT-2	6.19	78.54 _{1.44}
dDPO + dSFT	7.00	86.07 _{1.22}

- dDPO – dSFT: fine-tunes the base model directly with DPO for one epoch on UltraFeedback.
- dSFT-1: fine-tunes the base model with SFT for one epoch on UltraChat.
- dSFT-2: applies dSFT-1 first, followed by one more epoch of SFT on the top-ranked completions of UltraFeedback.
- dDPO + dSFT: applies dSFT-1 first, followed by one epoch of DPO on UltraFeedback.

Ablation study: Is Preference Optimization Necessary?

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Ablation study: Is Preference Optimization Necessary?



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- **dDPO + dSFT: applies dSFT-1 first, followed by one epoch of DPO on UltraFeedback.**

Backup Slides



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FLAN Task Clustering

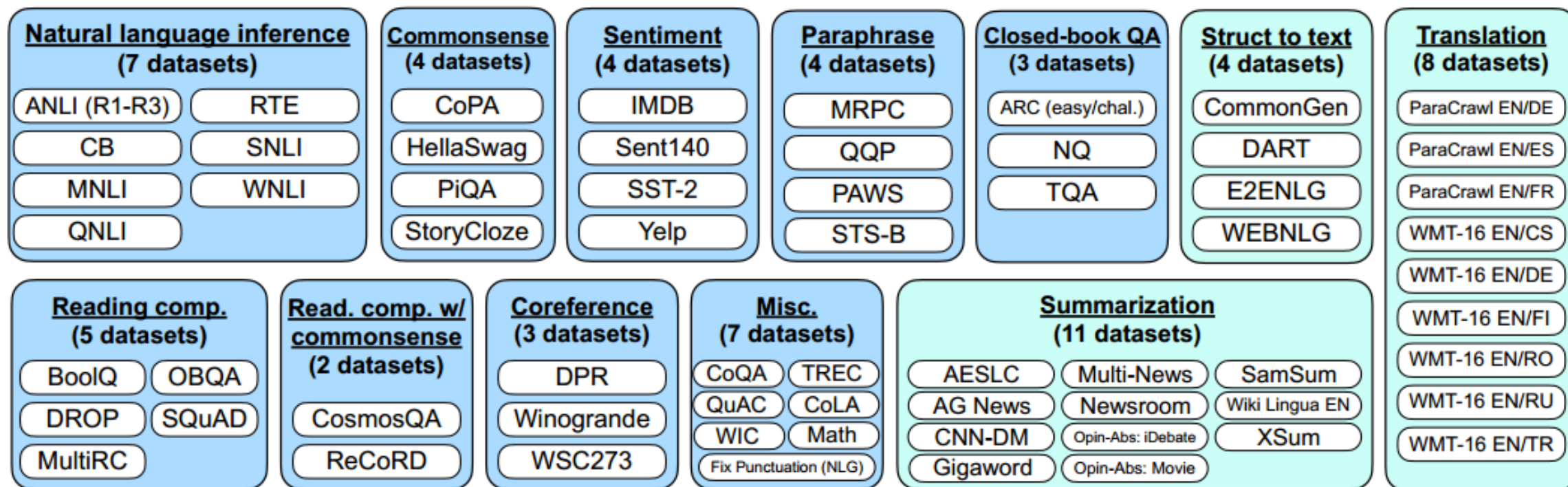
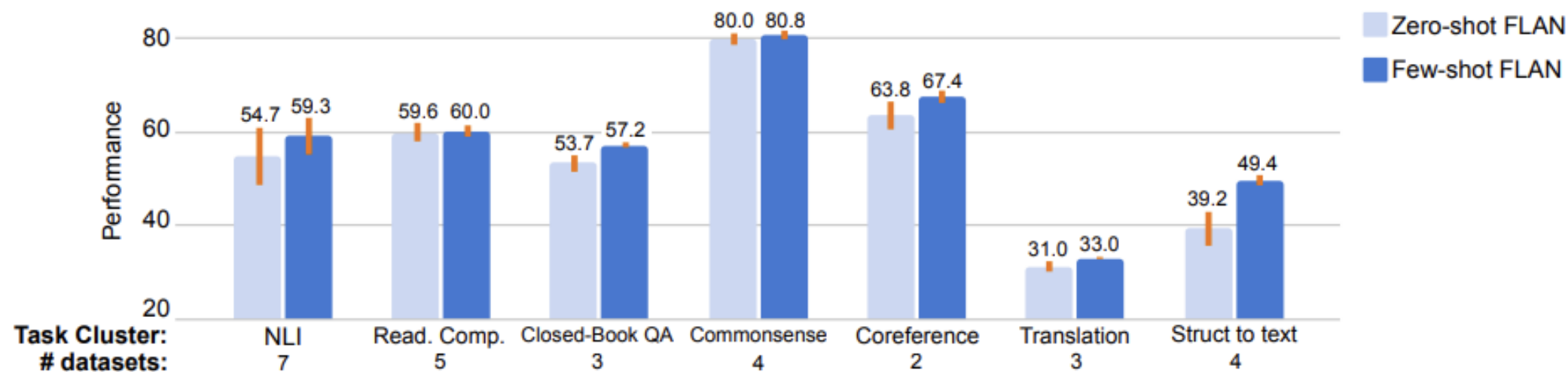


Figure 3: Datasets and task clusters used in this paper (NLU tasks in blue; NLG tasks in teal).

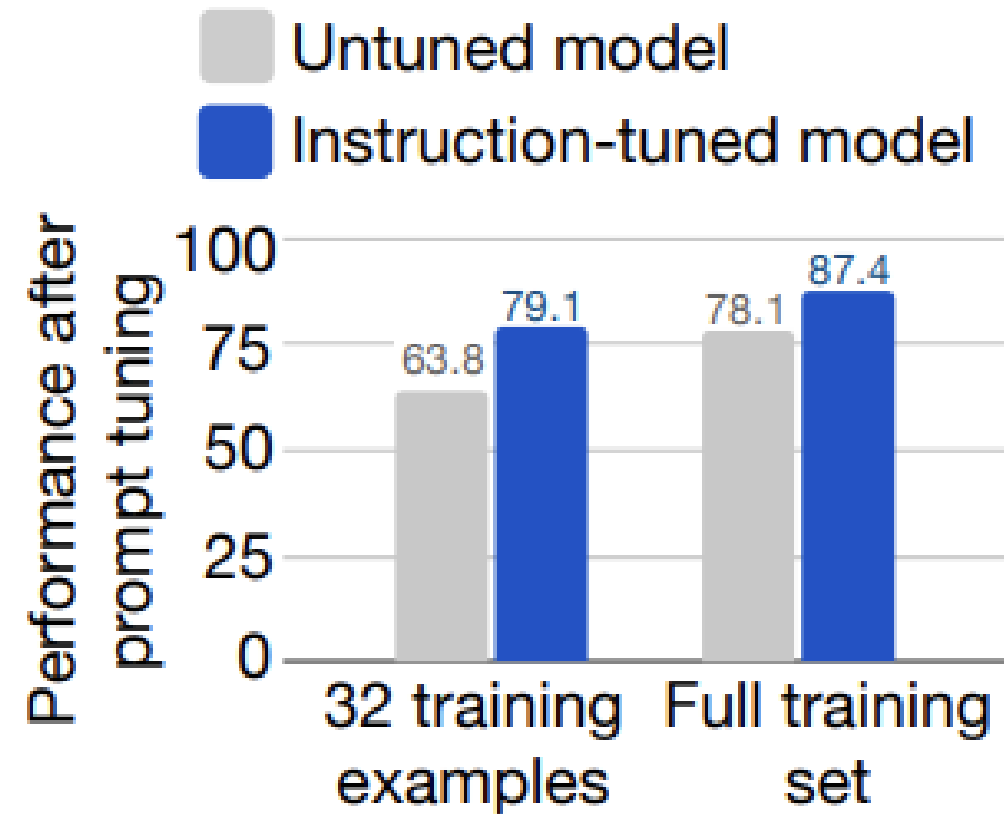
FLAN Ablation: Few-shot Instructions



FLAN with few-shot instructions still outperforms zero-shot FLAN

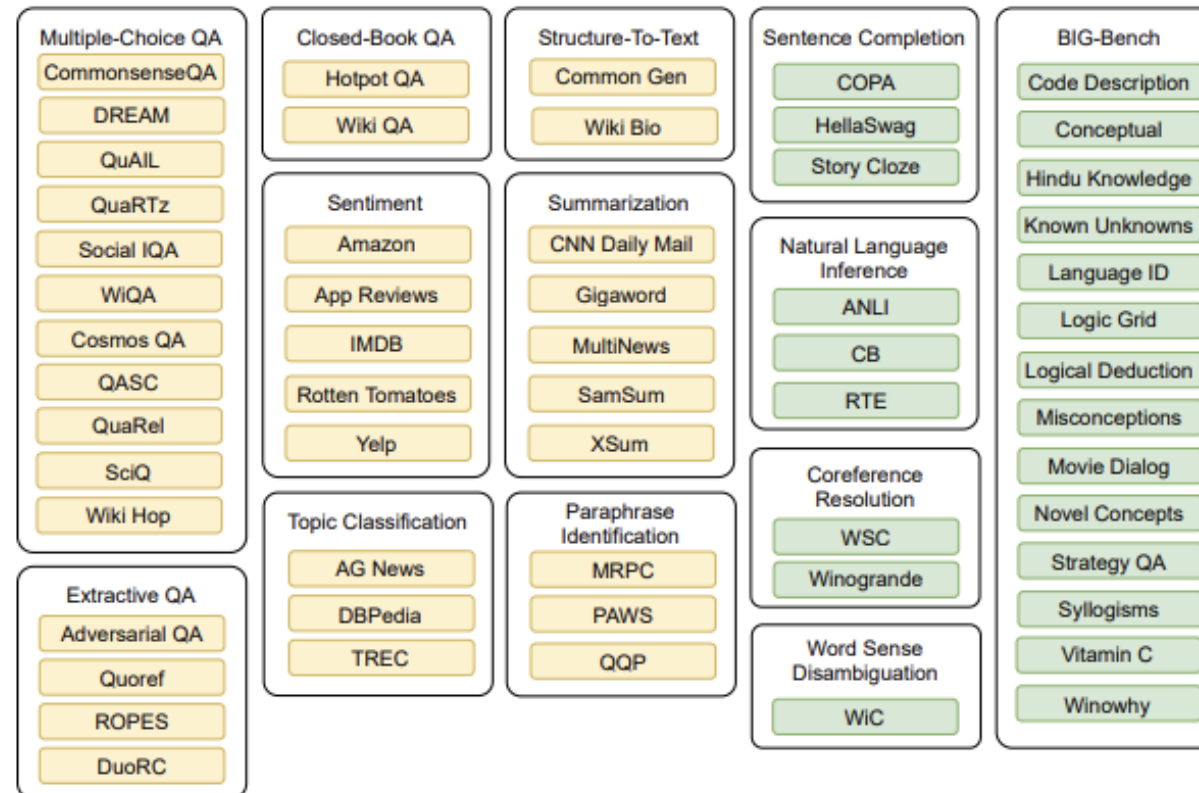
FLAN Ablation: Better Prompt Tuning

Instruction tuning helps
with prompt tuning
Effectiveness



Average performance on SuperGLUE dev set

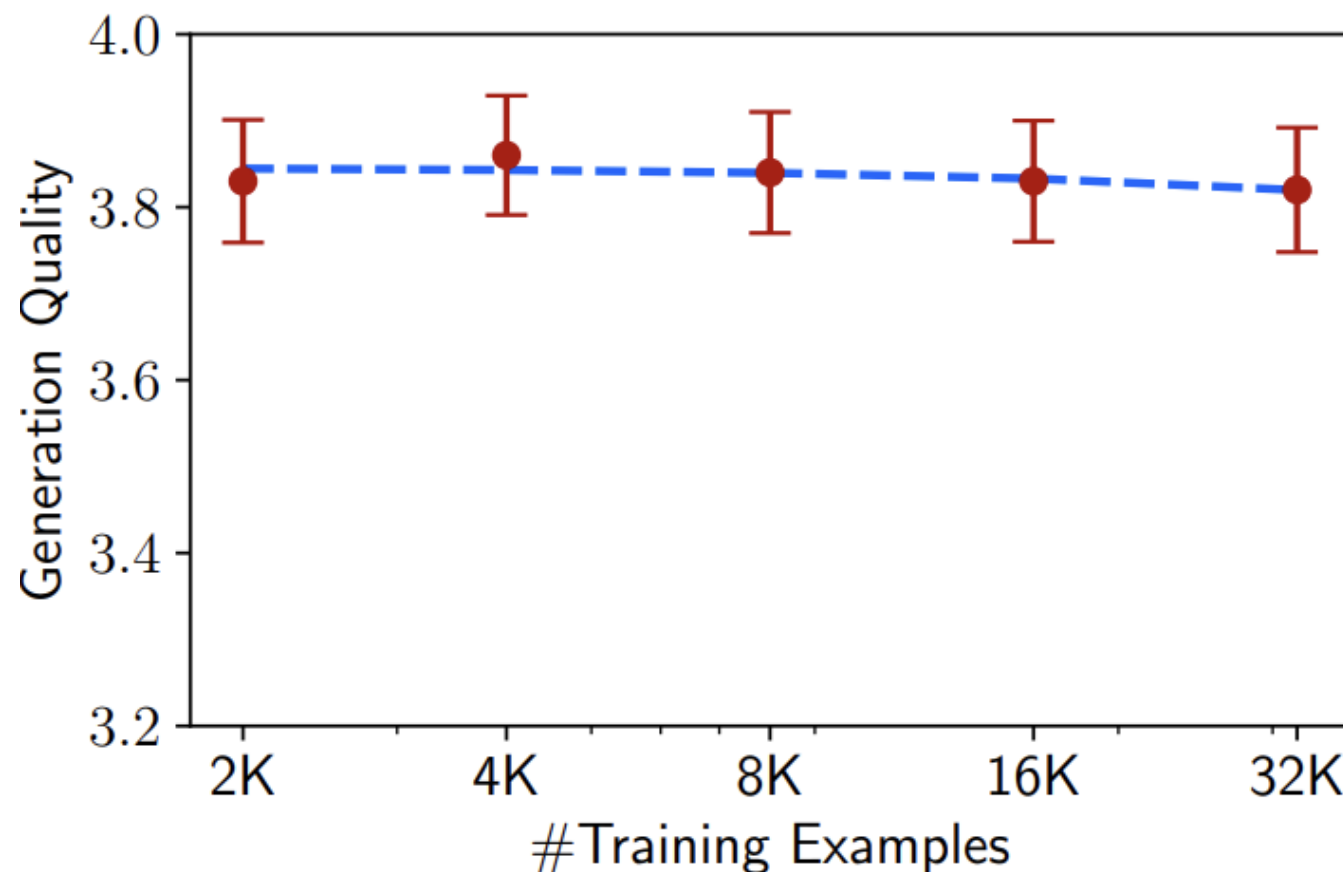
T0 Dataset Clustering into tasks



green dataset are held-out

LIMA Ablation study: Dataset Quantity

Increasing data quantity
doesn't further improve
the performance of the
7B model



LIMA Ablation study: Multi-turn



- In 6 out of 10 conversations, LIMA fails to follow the prompt within 3 interactions
- Added 30 multi-turn dialogue chains to

1000 training examples

- 10 Manually written
- 20 edited version of comment chains on stack exchange

Only 30 examples vastly improves multi-turn dialog performance of LIMA

